

Pyro Sensor
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FINAL REPORT

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PYRO SENSOR
OSCAR HERNANDEZ
ALFONSO WONG
MARTIN FORD

CONCEPT OF OPERATIONS

REVISION 3
10 APRIL 2025

CONCEPT OF OPERATIONS
FOR
Pyro Sensor

TEAM 24
PYRO SENSOR

APPROVED BY:

Alfonso Wong 04/10/2025

Prof. John Lusher 04/10/2025

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Change Record

Rev.	Date	Originator	Approvals	Description
1	09/15/2024	Pyro Sensor		Draft Release
2	02/11/2025	Pyro Sensor		Network Communication
3	04/10/2025	Pyro Sensor		First Final Revision Edit
4				

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Figure 1: Pyro Sensor Block Diagram

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1. Executive Summary

Fire and smoke detection in buildings is crucial for ensuring public safety. To improve on traditional systems, our project will develop a detection system that can instantly alert users and firefighters to fire or smoke events. The system will utilize room-specific sensors connected to a smartphone application, providing immediate notifications when an issue is detected. The app will offer information such as the precise location and duration of the fire or smoke, to the users and firefighters allowing for faster evacuations and emergency responses. The app will use Firebase which does real-time data communication, the system can be scalable which can be used for various types of buildings. Our sensor will be an advanced, reliable alternative to existing fire and smoke detection technologies, enhancing both safety and response times in critical situations.



2. Introduction

This document introduces the Pyro Sensor, a system capable of actively monitoring house fires and informing firefighters and residents. The Pyro Sensor will track the fire's location and start time, and notifications will be available through a mobile app.

2.1. Background

The current fire alarm systems utilized in most homes and buildings are different types of smoke detectors. The primary function of these are to detect and audibly warn of smoke and fire. However, only some systems include the capability to send a notification in an emergency. In fact, these detectors only send a notification to the manufacturer, which then has to be routed in a call to the fire department.

The Pyro Sensor addresses this issue as it both sets off an audible alarm and sends a notification through a mobile application directly to the fire department, therefore taking out the middleman. Additionally, the app sends a report to the fire department detailing time and location telemetry, providing more useful information than a basic alert.

2.2. Overview

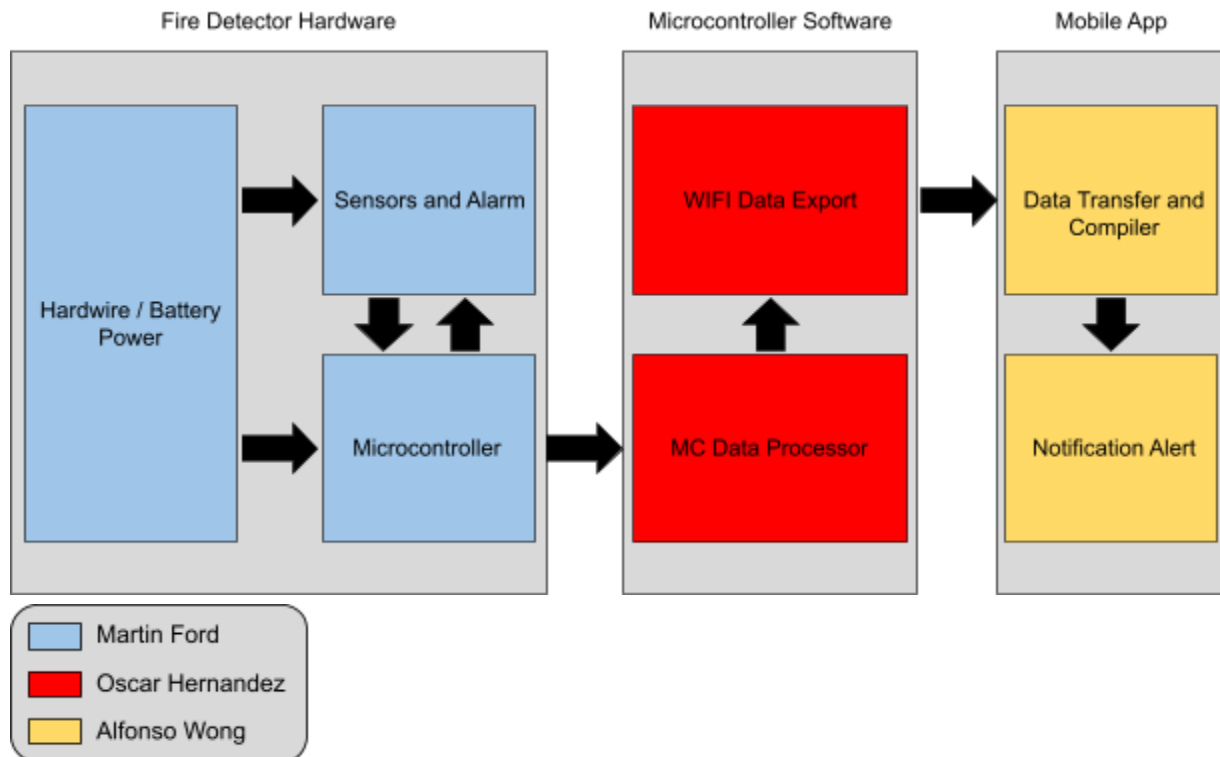


Figure 1: Pyro Sensor Block Diagram

Our system will be used to detect fire in a specific room in a building. Fire will be detected through a MQ2 gas sensor, UV and IR sensors, and a photometric smoke sensor. These sensors will relay their data to a microcontroller, which will process the data and determine if a fire has started. This hardware will be hardwired, but will also have a backup battery for power in the event of a power outage. Once a fire is detected, the microcontroller will send a signal back to sound an alarm through a speaker. The microcontroller will also send a notification to a mobile application when a fire starts and provide real-time timestamps to track the fire's growth in the building.

2.3. Referenced Documents and Standards

- [1] National Fire Protection Association, NFPA 72: National Fire Alarm and Signaling Code, 2019. <https://edufire.ir/storage/Library/elam/NFPA%2072-2019.pdf>
- [2] Detector Electronics Corporation, SmokeWatch™ U5015 Explosion Proof Smoke Detector Specifications, Rev. 2.2, 2020. <https://www.det-tronics.com/content/documents/U5015-Specifications-90-1217-2.2.pdf>
- [3] Espressif Systems, ESP-IDF Programming Guide for ESP32-S3, v5.4.1. <https://docs.espressif.com/projects/esp-idf/en/v5.4.1/esp32s3/index.html>
- [4] Analog Devices, ADPD188BI: Integrated Optical Module for Smoke Detection, Data Sheet. <https://www.analog.com/media/en/technical-documentation/data-sheets/adpd188bi.pdf>
- [5] Firebase Documentation. <https://firebase.google.com/docs>
- [6] Twilio Docs – Programmable Messaging API. <https://www.twilio.com/docs>
- [7] Android Developers. *Jetpack and AndroidX Libraries*. <https://developer.android.com/jetpack>
- [8] Android Developers. *Build UI with XML*. <https://developer.android.com/guide/topics/ui/declaring-layout>

3. Operating Concept

3.1. Scope

The Pyro Sensor will enable the fire department to be notified directly and immediately, through a central dispatcher. Additionally it will allow firefighters to receive real-time data consisting of time stamps and the location of fire in each room, allowing the firefighters to track the spread of the fire. Ultimately, this will improve the safety of both the firefighters and residents.

3.2. Operational Description and Constraints

For both commercial and residential buildings, the Pyro Sensor is meant to be placed in each room to enable more specific and accurate fire tracking.

The constraints the Pyro Sensor system will encounter include:

- The system must not trigger false alarms from non-fire related sources like steam or cooking smoke.
- The backup battery must be rated to last through a typical power outage, and the system must alert of low battery level.
- The microcontroller should maintain stable communication with the app, and quickly recover from signal loss.
- The app interface should be easy-to-follow, users should be able to track real time sensor status.
- The system should allow easy setup of new sensors through the app, these sensors should also integrate into the existing network.

3.3. System Description

3.3.1. Hardwire and Battery Power

This subsystem will supply power to the sensors, alarm and microcontroller. The primary power source will be hardwired electricity from the room that the Pyro Sensor is installed in. A battery will also be used as a secondary power source in case of a power outage. The hardwired power source will ensure a sufficient power supply to account for the high power consumption of the microcontroller due to utilizing Wi-Fi, while the battery will ensure the Pyro Sensor is never without power.

3.3.2. Sensors and Alarm

This subsystem consists of all the hardware peripherals that will communicate with the microcontroller. The sensors, including the photoelectric, ionization and UV/IR sensors will monitor the room it is stationed in, sending the data to the microcontroller to be processed. In the event of a fire, the sensors will continue to monitor the room and send data to the microcontroller for the real-time fire spread

report. Additionally, the alarm speaker will be set off and make an audible alarm sound.

3.3.3. Microcontroller Software and Network

The microcontroller firmware will manage Wi-Fi connectivity with the building router, receiving credentials via Bluetooth from the app. It will continuously monitor environmental changes for fire detection and handle Wi-Fi disconnections. The firmware should also support requests from the app.

3.3.4. Data Transfer and Compiler

The data from the microcontroller is received and processed. This includes saving it in cloud storage to have access to previous events. As well as data organization and presentation.

3.3.5. Notification Receiver

After the data is received, compiled, and organized, an alert notification will be sent out to both users and firefighters about the fire that has erupted. They will have access to information regarding the fire such as when the fire started, location, and current temperature levels.

3.4. *Modes of Operations*

The Pyro Sensor system has three modes: Safe, Warning, and Alarm. In Safe status, the system monitors environmental data and communicates the updates to the app. This mode also tests battery level, and Reset and Delete requests from the app. In Alarm status, the system triggers an alarm once a certain threshold is met, and sends a notification with detailed information to the app, including real-time updates of the fire's growth to the fire department and time-stamp of the fire. In Warning status, the app detects early hazards when at least one sensor exceeds Safe thresholds, but not enough to trigger the buzzer.

3.5. *Users*

The Pyro Sensor system is marketed towards local firefighting departments for use in residential households with an existing WiFi network. Residents require minimal training for daily use, as the app features a user-friendly interface. However, some training might be required when setting up the devices with the house wiring. Once installed, residents benefit from quicker evacuations and better emergency preparedness in the case of a fire event. Fire departments will need extra training to interpret app data and integrate it into response protocols, which will lead to faster fire suppression and improved safety.

3.6. *Support*

The Pyro Sensor package will include a manual with detailed information about the system: safety precautions, installation and usage instructions, and troubleshooting. Additionally, the

mobile app will include a support page which will include a frequent questions and answers section for the sensor.

4. Scenarios

4.1. *Residential Building*

4.1.1. House

In a standalone house, sensors are placed in key areas such as the kitchen, bedrooms, living spaces, and garages. When a fire is detected, the system immediately notifies the homeowners through the mobile app, even if they are away. The alert includes the fire's exact starting location and real-time updates on its progression, which can also be shared with local fire services to speed up emergency response and minimize property damage.

4.1.2. Apartment

In an apartment unit, sensors are typically installed in high-risk areas like kitchens and living rooms. Alerts are sent directly to the tenant's mobile app, providing real-time data and timestamps of the fire's behavior. Building management is also notified to coordinate evacuations and fire department communication, ensuring both the individual unit and neighboring units are protected quickly.

4.2. *Commercial Building*

4.2.1. Hotel

In a hotel, sensors are installed in guest rooms, hallways, lobbies, kitchens, and service areas. If a fire starts, alerts are sent to the hotel management and directly to a central fire control system, along with real-time reports. Mobile notifications also allow staff to assist in safe evacuation procedures and guide guests away from danger zones before firefighters arrive.

4.2.2. Business

In a business setting, such as a retail store or office space, sensors are installed in customer areas, storage rooms, and employee-only zones. When a fire is detected, real-time alerts are sent to management and security personnel through the app, helping coordinate rapid evacuations and system shutdowns if necessary. Fire department teams also receive detailed updates, improving the speed and effectiveness of their intervention.

5. Analysis

5.1. *Summary of Proposed Improvements*

- This system will have multiple types of sensors, allowing for accurate readings and detailed data.
- The microcontroller will send a notification directly to the fire department, without having to notify a manufacturer middleman.
- The microcontroller will send a real-time data report to the mobile application including timestamps, intensity and spread of the fire.

5.2. *Disadvantages and Limitations*

- Devices located far from the router may encounter slower connection response.
- A network must always be present and working for the devices to transmit data.
- The host microcontroller will slow down in performance under heavy load from multiple devices.

5.3. *Alternatives*

- Each device could be communicating directly with the router, instead of relying on a main microcontroller.
- Transmission between microcontrollers could be done through Bluetooth, and only the Host MC would use Wi-Fi.
- Individual sensors could be wired to a control panel that would set an alarm locally.

5.4. *Impact*

- Reduces environmental damage by providing early fire detection and preventing larger fires that could release harmful pollutants.
- Enhances public safety by allowing quicker fire responses and more efficient evacuations by the authorities, minimizing injuries and fatalities.
- Helps reduce property damage and the need for excessive firefighting resources, contributing to lower economic losses and recovery costs.
- Ensure privacy for users by not collecting unnecessary personal data from building occupants.

PYRO SENSOR
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FUNCTIONAL SYSTEM REQUIREMENTS

REVISION – Draft
24 September 2024

FUNCTIONAL SYSTEM REQUIREMENTS FOR Pyro Sensor

TEAM 24

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Change Record

Rev	Date	Originator	Approvals	Description
.				
-	9/24/2024	Martin Ford		Draft Release
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1. Introduction

1.1. Purpose and Scope

Detecting and tracking fires in infrastructure is integral to life preservation. We aim to create a system that utilizes a variety of sensors to detect building fires. The combination of multiple sensors and information processing software will reduce false positives and allow both fast detection and accurate tracking. The Pyro Sensor will notify and send a real-time tracking report to the fire department and residents in the event of a fire, which will be done by sending information to an app over WIFI. With our system having an emphasis on speed and accuracy of information delivery, the Pyro Sensor will increase the efficiency of the firefighters and the safety of the residents.

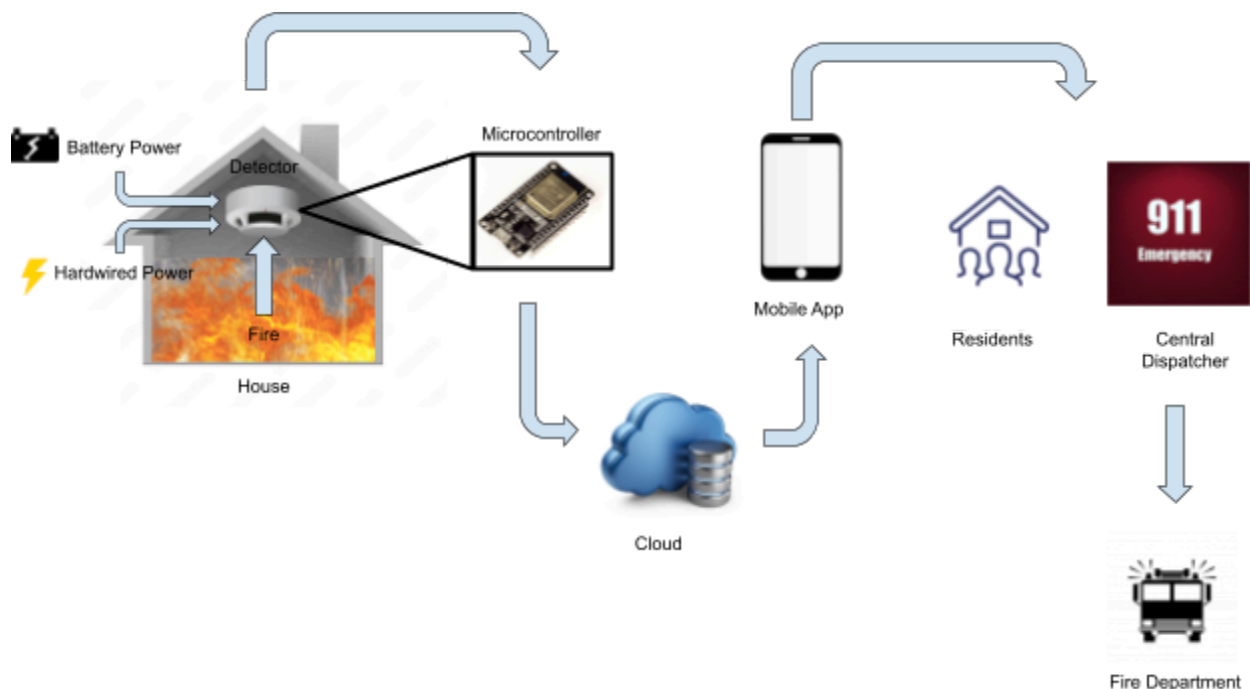


Figure 1. Project Conceptual Image

1.2. Responsibility and Change Authority

As the team leader, Alfonso Wong, holds the responsibility for ensuring that all project requirements are fully met. Any proposed changes to the project will be collectively reviewed and decided upon by the entire team, our sponsor, and Professor John Lusher II ensuring that all adjustments are carefully considered and aligned with the project's goals.

Subsystem	Responsibility
Fire Detector Hardware	Martin Ford
Microcontroller Software	Oscar Hernandez
Mobile App	Alfonso Wong

Table 1: Subsystem Leads

2. Applicable and Reference Documents

2.1. Applicable Documents

The following documents, of the exact issue and revision shown, form a part of this specification to the extent specified herein:

Document Number	Revision/Release Date	Document Title
IEEE 802.11n	2009	wireless local area networks
RFC 6864	February 2013	Internet Protocol (IPv4)
RFC 9293	August 2022	Transmission Control Protocol (TCP)
IEEE 829	July 2008	Standard for Software and System Test Documentation

Table 2: Applicable Documents

2.2. Reference Documents

The following documents are reference documents utilized in the development of this specification. These documents do not form a part of this specification and are not controlled by their reference herein.

Document Number	Revision/Release Date	Document Title
IEEE 802.15.4	2020	low-rate wireless personal area network (LR-WPAN)
ISO 27001	October 2022	Information security, cybersecurity and privacy protection — Information security management systems — Requirements

Table 3: Reference Documents

2.3. *Order of Precedence*

In the event of a conflict between our project specification and **IEEE 829**, which requires a formalized and detailed testing process, the project spec takes precedence. Due to the strict timeline and the need to deploy the system for testing, our team has decided on a more streamlined testing process to ensure that real-time alert performance is tested early in the development cycle.

3. Requirements

3.1. System Definition

The Pyro Sensor is a fast and accurate alarm system aimed at detecting and notifying relevant personnel of a building fire. The system is built to alert residents and instantly notify and continually report useful information to firefighters during a fire. The Pyro Sensor has three subsystems: Fire Detector Hardware, Microcontroller Software, and Mobile App.

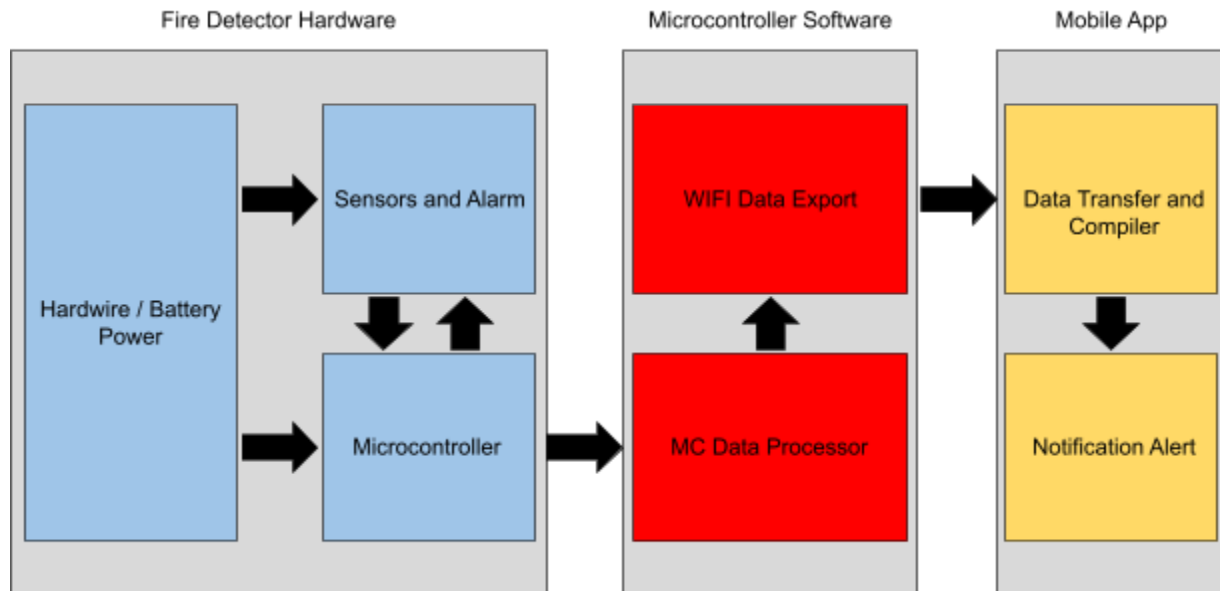


Figure 2. Block Diagram of System

The fire detector will have both hardwired power from the room it is installed in and a backup battery as a failsafe to power the sensors, alarm audio system and microcontroller. The sensors will detect a fire in the room and send signals to the microcontroller. The microcontroller will process the information and determine if a fire is detected. In the event of a fire, the microcontroller will send a signal back to the main board to set off the audible alarm. Additionally, the microcontroller will send an alert signal via Wi-Fi to the cloud, where it will be stored and then pulled by the mobile app from the cloud. Once the mobile app has received the alert signal, the app will notify the fire department and residents of the fire. After the initial fire notification, the sensors will continue to send data to the microcontroller to be processed and sent to the mobile app via the cloud as a real-time fire-tracking report for the firefighters.

3.2. Characteristics

3.2.1. Functional / Performance Requirements

3.2.1.1. Fire Detection Accuracy

The fire detection system shall maintain a detection accuracy of over 95%.

Rationale: The system must maintain a low false negative rate, but must account for small variations in current to sensors in the case of low battery charge.

3.2.1.2. False Alarm Rate

The fire detection algorithm shall maintain a false alarm rate of less than 5%.

Rationale: The system must consider detection of benign small particles such as dust or steam. A low false positive rate ensures that the system remains reliable and avoids unnecessary alarms that can lead to alarm fatigue or burden emergency services. Requiring activation of multiple sensors within a 15 to 30 second window helps filter out false alarms while quickly reporting real fires.

3.2.1.3. Backup Battery Operating Time

The fire detection hardware shall be able to function on battery power for 7 days.

Rationale: The fire detection hardware will primarily run on hardwired power from the building at all times. However, the system needs to be able to run on battery power for a sufficient amount of time to account for an extended power outage.

3.2.1.4. Network Transmission Speed

Sensor data shall be aggregated from the microcontroller, then forwarded to the router, with a delay of less than 10 seconds and a packet delivery success rate of 98%.

Rationale: Reducing delays in collecting data from multiple sensors and transmitting it to the router allows for a quicker response from firefighters, thus minimizing the spread of the fire and damage done to property.

3.2.1.5. Fire Hazard Response Time

Assuming uninterrupted communication channels, emergency services and building occupants shall receive a notification within 30 seconds of a fire, and a sound alarm will activate.

Rationale: A total delay of 30 seconds improves the chances of successful evacuation, and effective hazard control before the fire engulfs the area.

3.2.1.6. App compatibility

The mobile app shall be compatible with Android devices to ensure accessibility for a broad range of users.

Rationale: Android holds a significant market share in mobile devices, so ensuring compatibility with Android systems increases the user base and accessibility of the fire detection system.

3.2.1.7. Smart-Diagnostics

Each device shall perform firmware and peripherals periodically and report system status to the app.

Rationale: Periodic system monitoring ensures sensors, communication modules, and backup battery will function properly during active fire events.

3.2.1.8. User Creation

The application must have the capability of letting users create an account which they will need to create an email and password.

Rationale: Having the ability to create an account for mobile applications is crucial for handling user data.

3.2.1.9. Application Notification System

The application must be able to send a notification to the user incase of an alarm trigger. A text notification will also be sent to a central dispatcher which will then alert the fire department.

Rationale: The most crucial part of this project is the quickness of the user and authorities being notified of a fire.

3.2.1.10. Functioning UI

It is essential for the application to have a functioning User Interface as a properly working UI is needed for an application to be usable.

Rationale: A working UI is critical for any type of application. If the UI is not usable, the user will not want to use the application.

3.2.1.11. Integrated Cloud Service

The mobile application will have the ability to pull and write data from cloud service and will be able to display it.

Rationale: The application needs to be able to show any data from the sensors and be able to display it on the app for users to see.

3.2.1.12. Sensor and Application Initial Connection

The initial setup between the application and the sensor will be through the use of bluetooth. The sensor will be linked with the user that added it.

Rationale: The initial connection can be tricky, but by using bluetooth, an efficient and fast connection can be made to make the initial setup quicker.

3.2.1.13. Sensor and Application Real-time Communication

The application and the sensor will communicate in real-time. This will be done through the use of Firebase Real-Time database. Testing will be done to ensure the application is able to display the information being provided to the database by the sensor with little to no lag.

Rationale: For quick notification incase of a fire, real-time communication between the sensor, database, and the mobile application is crucial.

3.2.1.14. Warning Signal

MCU sends warning signal to app for malfunctioning node.

Rationale: The device should function under intended design, such as ensuring the supply voltage is enough to power all its components. Otherwise, unpredictable behavior may occur.

3.2.2. Physical Characteristics

3.2.2.1. Device Mass

The mass of the fire detection hardware shall be less than or equal to 0.7 pounds.

Rationale: This is a characteristic consistent with most smoke alarms, making it easy to install the module in the ceiling socket.

3.2.2.2. Volume Envelope

The volume envelope of the fire detection hardware shall be less than or equal to 6 inches in diameter, and 3 inches in height.

Rationale: The diameter parameter is a requirement pertaining to the ceiling socket that the device will be mounted into. The height parameter is a characteristic consistent with most smoke alarms, dedicating an inch to the socket insertion portion and the rest protruding from the ceiling.

3.2.2.3. Installation

The installation information for the Pyro Sensor shall be provided to the customer through a user manual.

Rationale: This ensures that the customer has clear, detailed, and accessible instructions on how to correctly install and operate the sensor.

3.2.2.4. Mounting

The mounting information for the fire detection hardware shall be captured in the Pyro Sensor ICD.

Rationale: As the Pyro Sensor mounts to a standard socket in a building, the interface between the two includes mechanical and electrical details.

3.2.2.5. Power Source

The fire detection hardware shall be powered primarily through standard 120V hardwired power from the building, as well as a backup 9V battery.

Rationale: Using hardwired power will ensure long lasting functionality with minimal maintenance. The presence of a backup battery at the specified voltage will ensure ongoing functionality with sufficient power during a power outage.

3.2.2.6. Alarm Sound

The alarm audio speaker shall emit sound with a threshold of at least 65 to 85 dB from 10 ft away.

Rationale: This requirement is consistent with standard smoke alarm regulations specified by the NFPA, ensuring the alarm can be easily heard over an average noise level.

3.2.2.7. Wireless Range

The effective wireless coverage area of each detector shall be at least 31,416 square feet indoors.

Rationale: This coverage area is based on typical Wi-Fi 802.11n standard setting. Theoretically, the indoor range is about 230 feet, but interference from walls and other obstacles reduces this distance to approximately 100 feet, much less than the space between rooms.

3.2.3. Electrical Characteristics

3.2.3.1. Inputs

- a. The presence or absence of any combination of the input signals in accordance with ICD specifications applied in any sequence shall not damage the Pyro Sensor, reduce its life expectancy, or cause any malfunction, either when the unit is powered or when it is not.
- b. No sequence of command shall damage the Pyro Sensor, reduce its life expectancy, or cause any malfunction.

Rationale: By design, should limit the chance of damage or malfunction by user/technician error.

3.2.3.1.1 Power Consumption

The maximum peak power of the system shall not exceed 3 watts.

Rationale: This characteristic accounts for the many components and peripherals in the device that would be used at the same time during a fire including the sensors, alarm audio speaker, and microcontroller, which will be actively using WIFI, resulting in a higher power draw.

3.2.3.1.2 Input Voltage Level

The input voltage level for the Pyro Sensor shall be 5V.

Rationale: Most of our internal devices and peripherals are designed to run off of 5V, with the exception of a couple parts which will use a regulator.

3.2.3.2. Outputs

3.2.3.2.1 Data Output

The firefighter department shall have a view and status of all the sensors in the buildings with our devices installed.

Rationale: This information is given straight to the department to make quicker response times for fire emergencies.

3.2.3.2.2 Diagnostic Output

The Search and Rescue System shall include a diagnostic interface for control and data logging.

Rationale: Provides the ability to control things for debugging manually and a way to view/download the node map with associated potential targets.

3.2.3.3. Connectors

The Pyro Sensor shall use a standard pigtail wire set to hardwire to a building room and comply with NEC NFPA 70 Article 760.

Rationale: Conform to NFPA fire alarm standard.

3.2.3.4. Wiring

The Pyro Sensor shall follow the guidelines outlined in the NEC NFPA 70 Article 760.

Rationale: Conform to NFPA fire alarm standard.

3.2.4. Environmental Requirements

The Pyro Sensor shall be designed to withstand and operate in the environments and laboratory tests specified in the following section.

Rationale: The Pyro Sensor will need to function in a standard indoor setting.

3.2.4.1. Pressure (Altitude)

The Pyro Sensor shall be able to function properly at altitudes less than 3,000 ft.

Rationale: The circuitry, including primarily the ionization sensor, require staying below this altitude to function correctly.

3.2.4.2. Thermal

The fire detection hardware shall be installed in a room that ranges between 40° to 100° Fahrenheit under normal conditions.

Rationale: This requirement ensures the device will perform as intended according to the standard calibrations of the sensors.

3.2.5. Failure Propagation

The Pyro Sensor system shall not allow propagation of faults beyond the physical device.

Rationale: This prevents failures from escalating into larger issues that could compromise the functionality or safety of the entire system.

3.2.5.1. Failure Detection, Isolation, and Recovery (FDIR)

The Pyro Sensor system conducts a fault detection check continuously. Each node monitors battery levels, communication failures, and sensor malfunctions. During each check, a node will read its backup battery voltage level, alarm and sensors' status. If an issue is detected, a Warning signal with the node's location is sent to the database, which forwards it to the app for physical inspection.

Rationale: This requirement maintains a robust, safety-critical fire detection system.

3.2.5.1.1 Built In Test (BIT)

The Pyro Sensor system shall have an internal algorithm that will generate test signals and evaluate the sensor node responses and determine if there is a failure.

3.2.5.1.1.1 BIT Critical Fault Detection

The BIT shall be able to detect a critical fault in the Pyro Sensor system 90 percent of the time.

Rationale: This requirement ensures that, in the event of a fire, only a minimal number of alarms may fail due to pre-existing malfunctions..

3.2.5.1.1.2 BIT False Alarms

The BIT shall have a false alarm rate of less than 10 percent.

Rationale: This prevents unnecessary disruptions from verifying false alarms.

3.2.5.1.1.3 BIT Log

The BIT shall save the results of each test to a log that shall be stored in the mobile application.

Rationale: Saved records eases troubleshooting future problems, and opens up accountability from users.

3.2.5.1.2 Isolation and Recovery

The Pyro Sensor system should provide for fault isolation and recovery by enabling automatic detection, attempt at resetting faulty components, and sending a trouble signal notification.

Rationale: This ensures continuous operation, with the system handling minor issues, while user intervention is required, otherwise.

4. Support Requirements

The Pyro Sensor system requires an Android device with internet connection to access the app. The building must provide power to all detection nodes and have Wi-Fi for notification delivery.

Appendix A: Acronyms and Abbreviations

BIT	Built-In Test
Hz	Hertz
ICD	Interface Control Document
W	Watt
Hz	Hertz
V	Volts
dB	decibels

Appendix B: Definition of Terms

PYRO SENSOR
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INTERFACE CONTROL DOCUMENT

REVISION – Draft
26 September 2024

INTERFACE CONTROL DOCUMENT FOR PYRO SENSOR

Team 24

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Change Record

Rev	Date	Originator	Approvals	Description
.				
-	09/26/2024	Alfonso Wong		Draft Release

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1. Overview

The purpose of this document is to outline the details of how the fire detection sensors alert building occupants and fire department of a fire event. It will specify all the hardware of the Pyro Sensor, and how the peripheral signals are received by the microcontroller. The communication process between the microcontroller and the local router will also be provided. Finally, the transmission of the aggregated data from the microcontroller to the mobile application will be explained.

2. References and Definitions

2.1. References

Refer to section 2.2 of the Functional System Requirements document

2.2. Definitions

BIT	Built-In Test
Hz	Hertz
ICD	Interface Control Document
W	Watt
Hz	Hertz
V	Volts
dB	decibels
TBD	To Be Determined
WSN	Wireless Sensor Network
GHz	Gigahertz
oz	Ounces

3. Physical Interface

3.1. Weight

3.1.1 Main Control Unit

Component	Weight	Number of Items	Total Weight
ESP32-S3-WROOM-1	N/A	1	N/A
9V Battery	1.6oz	1	1.6oz

Table 1: Main Control Weight

3.1.2 Peripherals

Component	Weight	Number of Items	Total Weight
MQ-2	N/A	1	N/A
MIKROE-4299	0.75oz	1	0.75oz
Gikfun-EK8443 IR Sensor	0.28oz	1	0.28oz
GUVA-S12SD UV Sensor	N/A	1	N/A
PUI Audio Magnetic Buzzer	0.04oz	1	0.04oz

Table 2: Sensors and Buzzer

3.2. Dimensions

Dimensions are in inches

3.2.1. Dimension of Fire Detector Hardware Subsystem

3.2.2.

Component	Length	Width	Height
ESP32-S3-WROOM-1	1	0.709	0.122
9V Battery	1	0.6	1.9
MQ-2	0.787	0.787	0.906
MIKROE-4299	2.25	1	1
Gikfun-EK8443 IR Sensor	0.197	0.197	2.1
GUVA-S12SD UV Sensor	0.138	0.11	0.075
PUI Audio Magnetic Buzzer	0.378	0.378	0.413

3.3. Mounting Locations

Fire detectors should be placed in locations that adhere to local codes and regulations. Ideally, they should be installed near the center of the ceiling in each room, using available electrical ceiling boxes or sockets. This placement should be consistent across all rooms, avoiding areas near objects that may trigger false alarms.



4. Thermal Interface

The Pyro Sensor will not incorporate a thermal interface as the ESP32 hardware is well equipped to withstand the temperatures that the fire detection hardware module is designed for.

5. Electrical Interface

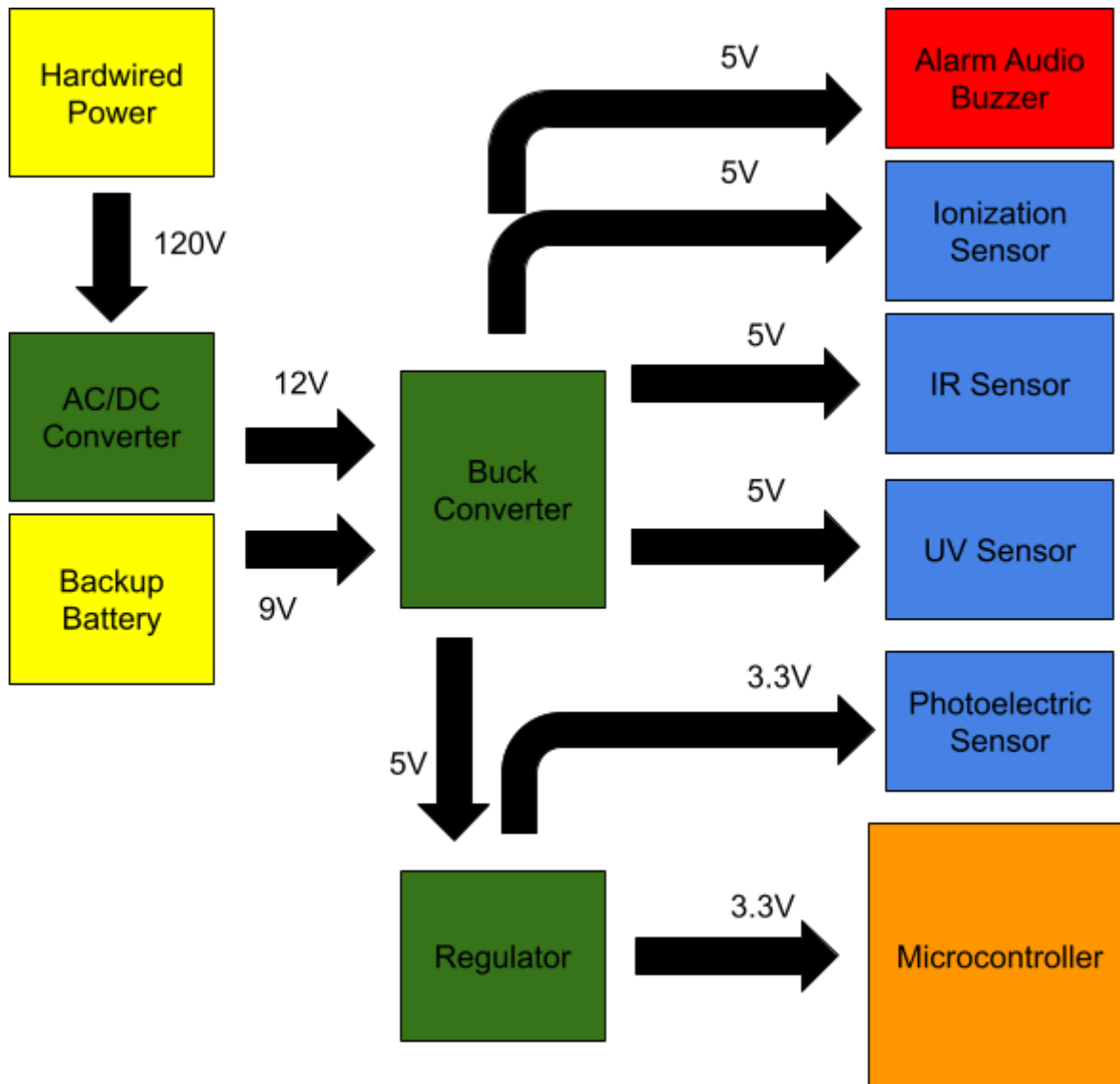


Figure 2: Electrical Interface Diagram

5.1. Primary Input Power

The fire detection hardware will be primarily powered by being hardwired to the room it is installed in, providing 120VAC. It will also have a 9VDC backup battery in case of a power outage.

5.2. Voltage and Current Levels

5.2.1. Maximum Values

Component	Voltage (V)	Current (mA)	Power (mW)
ESP32-S3-WROOM-1	3.6	240	864
MQ-2	5	150	750
MIKROE-4299	3.3	N/A	N/A
Gikfun-EK8443 IR Sensor	2.3V	30	70
GUVA-S12SD UV Sensor	5	1	5
PUI Audio Magnetic Buzzer	7	30	210

5.3. Signal Interfaces

5.3.1. Analog to Digital Converter

The MQ2 gas sensor, infrared sensor, ultraviolet sensor, and 9V battery all provide analog output voltages. To ensure these voltages are within range, voltage dividers are used to reduce them to a maximum of 3.3 V, the rated GPIO voltage for the ESP32-S3 MCU. The ESP32 then utilizes its ADC units with 12 dB attenuation, allowing it to accurately read GPIO voltages up to 3.1 V.

5.3.2. Inter-Integrated Circuit (I2C)

The smoke sensor module uses the I2C protocol to communicate with master devices. The MCU sets the clock speed to 100 kHz for the SDA and SCL lines, which are used to read and write data to the module registers to retrieve smoke levels.

5.3.3. Transistor-driven buzzer control

The MCU controls an active buzzer through a 5V powered transistor, switching it on and off. The buzzer sounds when the MCU biases the transistor into saturation, and remains off when the transistor is in cutoff.

5.4. User Control Interface

Users will have the ability to add or remove devices from their accounts and monitor the status of each device in real-time. They will also have full control over push notification settings. Additionally, users can grant authorized individuals access to view the status of

their devices. The fire department will be granted restricted access, allowing them to monitor the status of all devices within their jurisdiction.

6. Communications / Device Interface Protocols

6.1. *Wireless Communications*

6.1.1. Bluetooth Low Energy

During boot-up, the MCU enables BLE as a GAP server to connect with the client app, exposing public GATT characteristics for the app to send Wi-Fi credentials and device location.

6.1.2. Wi-Fi

The ESP32-S3-WROOM-1 microcontroller has a built-in antenna that supports the IEEE 802.11 b/g/n standard at 2.4 GHz. This module will transmit MCU data to the building's router, and from there, to the cloud server using the TCP/IP protocol.

6.2. *Host Device*

6.2.1 Pyro Sensor node

The Pyro Sensor device will merge signals from all of its sensors. It will filter out and categorize data, and transmit it to the cloud.

6.3. *Device Peripheral Interface*

The sensors and alarm will connect with the ESP32-S3 via GPIO pins, enabling the microcontroller to process changes in input signals and output a sound when necessary.

6.4. *Mobile App Communication*

The mobile app will receive data from the cloud by pulling real-time alerts and device status updates from Firebase, a cloud service platform. Firebase facilitates seamless communication between the sensors and the app, ensuring that notifications and device data are delivered promptly over Wi-Fi.

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MILESTONES AND VALIDATION PLAN

REVISION – Final Draft
26 September 2025

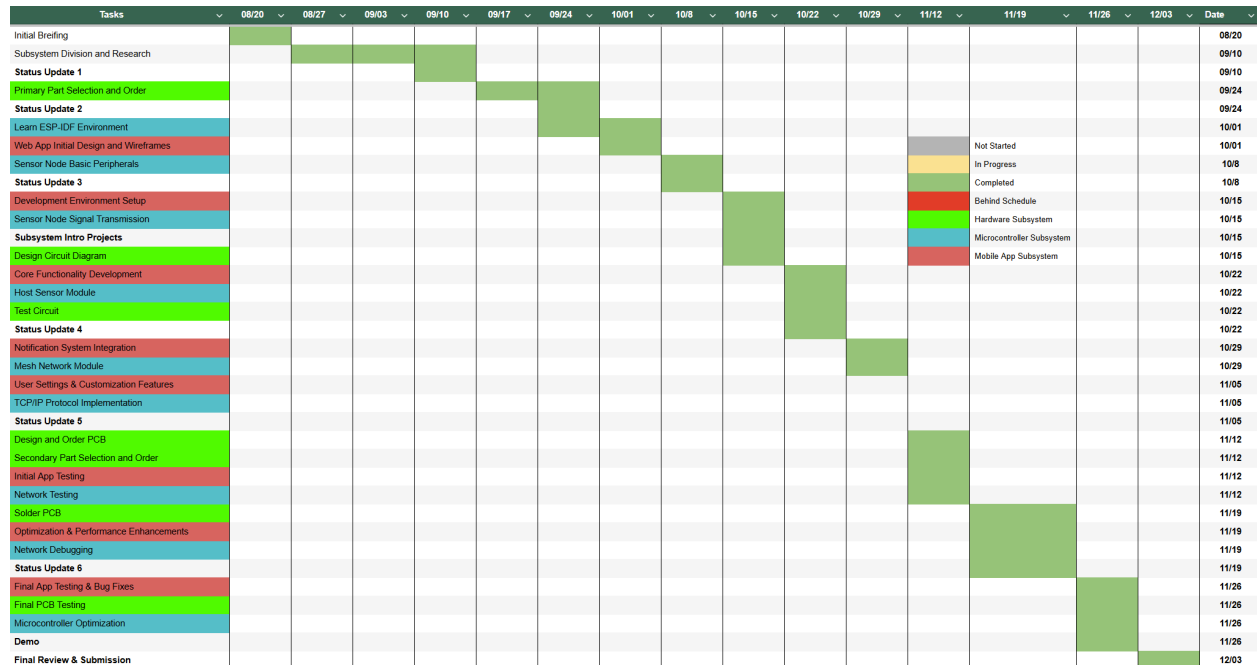


Figure 1: ECEN 403 Schedule

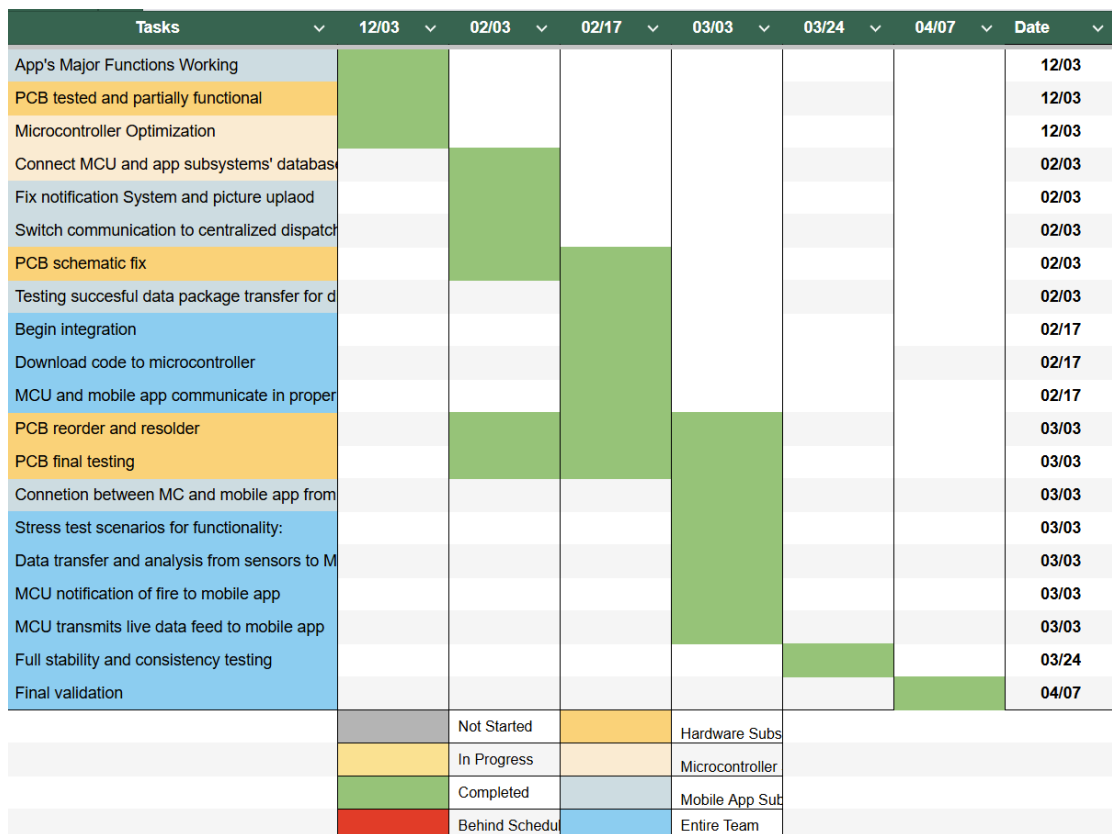


Figure 2: ECEN 404 Schedule

Test Name	Successful Criteria	Result
Sensor Functionality	Ionization and photoelectric sensors activate on smoke contact and output signals. IR and UV sensors detect flames up to 10 ft away and output signals.	Successful
Speaker Functionality	Speaker emits sound from signal at >65 dB.	Successful
Operating Time	System operates and receives full power continuously for 1 hour on each power source.	Successful
Network	MCU is able to aggregate data from sensors to router.	Successful
Dimensions	Mass: System (including casing) does not exceed 0.7lbs. Volume Envelope: System (including casing) does not exceed 6.5" diameter and 3" in height. Mounting: System casing successfully inserts and holds in ceiling smoke alarm slot.	Successful
Voltage Levels	Peripherals: Ionization, IR and UV sensors each receive 5V; audio buzzer receives 5V; photoelectric sensor receives 3.3V. Microcontroller receives 3.3V.	Successful
Thermal Resistance	System operates successfully in environment within temperatures ranging from 40F to 120F.	Successful
Trouble Signal	MCU sends trouble signal to app for malfunctions.	Successful
User Profile Creation	Users are able to create an account and be able to sign in.	Successful
Push Notifications	Be able to push notifications to users and send text to the central dispatcher.	Successful
Design UI	Have a working UI and test with an emulator.	Successful

Integrate Cloud Service	Mobile app has ability to pull data from cloud database and displays it.	Successful
App compatible on Android devices	App can be used on most Android devices	Successful
MCU Peripherals Interface	MCU reads signals from all sensors, and activates/deactivates alarms.	Successful
Device and Application Communication	A new device can be added to the user profile in application.	Successful
Real Time Communication	MCU is feeding constant information to the database and it is being displayed in real time in mobile application.	Successful
Trigger Warning from MC	MCU must send a trigger alert of ongoing event and application must send notification to user and dispatch	Successful
Early Fire Detection	System detects smoke/flame with 95% accuracy. Detected alarm activates buzzer.	Successful

PYRO SENSOR
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SUBSYSTEM REPORTS

REVISION – 2
05 December 2024

PYRO SENSOR

Martin Ford

HARDWARE

REVISION – Final Report
05 December 2024

Introduction

The hardware subsystem is designed to utilize a primary and secondary power source to power and utilize a variety of sensors. These sensors detect fires and send signals to the microcontroller to be processed and to determine if there is a fire. The hardware subsystem was tested to make sure that the full circuit was successfully powered, that the sensors would deliver signals as intended, and that the alarm speaker system worked as intended.

Details

A block diagram of the subsystem is shown below:

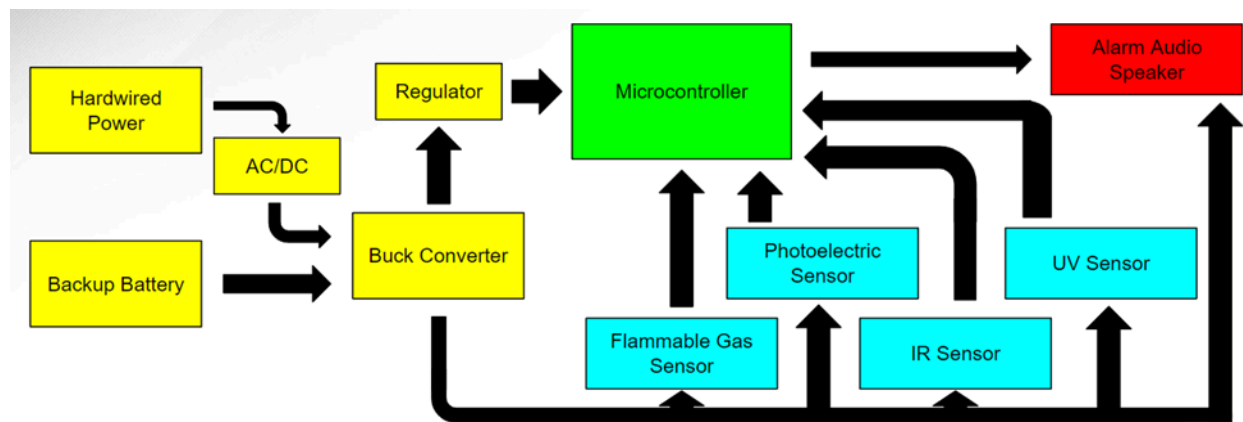


Figure 1: Functional Block Diagram of the Hardware Subsystem

The first and primary element of the design is the sensors. The Pyro Sensor hardware utilizes four sensors carefully chosen to detect and track fires. Firstly, the MIKROE-4299 photoelectric sensor and the MQ-2 gas sensor were chosen to work in tandem to quickly detect and notify of an initial fire. Both sensors work by detecting smoke using different methods, mitigating the risk of false positives. Similarly, the GUVVA-S12SD ultraviolet sensor and the Gikfun-EK8443 infrared sensor were chosen to work in tandem to track the spread of fire after it has already been detected. Both sensors work as photodiode receivers that generate signal currents and are compared to generate more accurate results. The photoelectric sensor was bought as a prebuilt module, whereas the other sensors were each bought as barebones components with custom designed modules in the circuit schematic. Each module is primarily composed of a comparator, which would compare the output signal current or voltage from the sensor to the 5V being distributed throughout the circuit reduced to an appropriate reference voltage. This would amplify the signal of the sensor into a usable signal sent to the microcontroller. Each module also incorporates capacitors to reduce noise.

The second element of the design is the buzzer. The PUI Audio Magnetic Buzzer was bought with compact size, low power draw, and high decibel volume in mind. Once a fire is detected, the microcontroller sends a control signal to a transistor to allow the buzzer to receive a 5V signal and emit a local, audible alarm.

The third element of the design is the microcontroller, which receives both power and signals from the circuit. Once it receives the signals from the sensors and processes the data, it sends a signal to the buzzer as an alarm.

Finally, the fourth element of the design is the power distribution system. This consists of a primary and secondary power source, which are a 120VAC source from the room the Pyro Sensor module is installed in and a 9VDC battery respectively. The 120VAC is run through an AC/DC converter to become 12VDC. The resulting 12V primary source and 9V secondary source are joined in a diode-or configuration that enables the primary source to power the circuit unless there is a power outage, in which case the secondary source will take over. The dominant power source is sent through a buck converter, outputting a stable 5V to the rest of the circuit, powering the microcontroller, sensors, comparators for the sensors, and the buzzer. For the microcontroller and photoelectric sensor specifically, the 5V is sent through a regulator to output a stable 3.3V. Additionally, the battery is connected to the microcontroller through a voltage divider to monitor its capacity. All these listed connections can be seen in the block diagram above and the circuit schematic below.

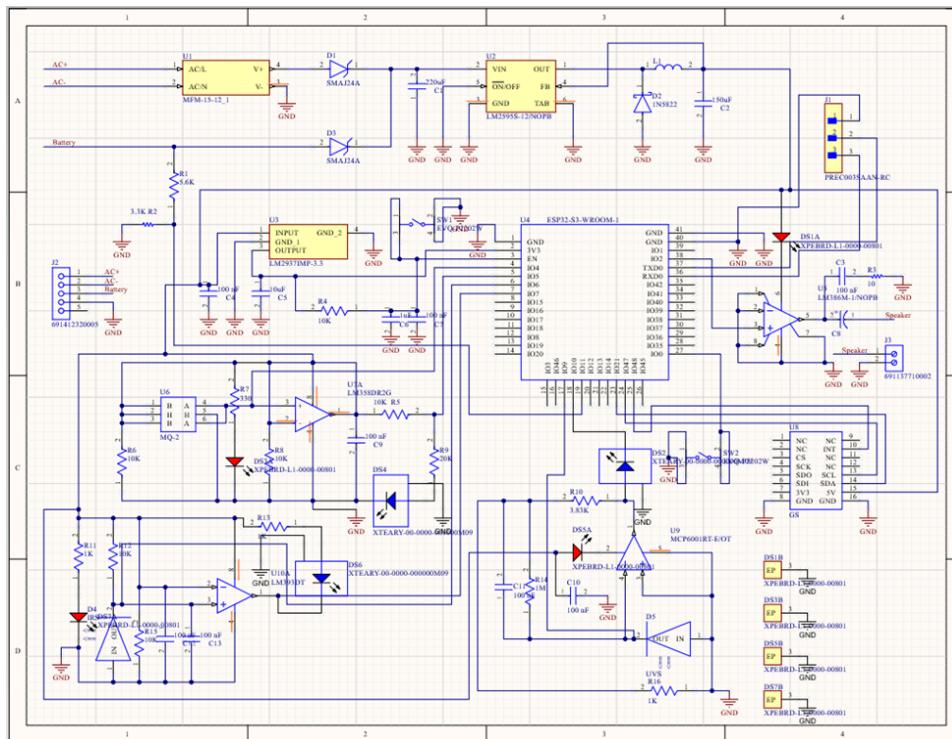


Figure 2: Circuit Schematic of the Hardware Subsystem

The PCB was designed with compactness in mind as the whole board is only 4" x 4" so that it can fit in a standard smoke detector plastic housing. Additionally, the traces were all designed with a width of 10 mils given that all currents going through the circuit will stay below 1A.

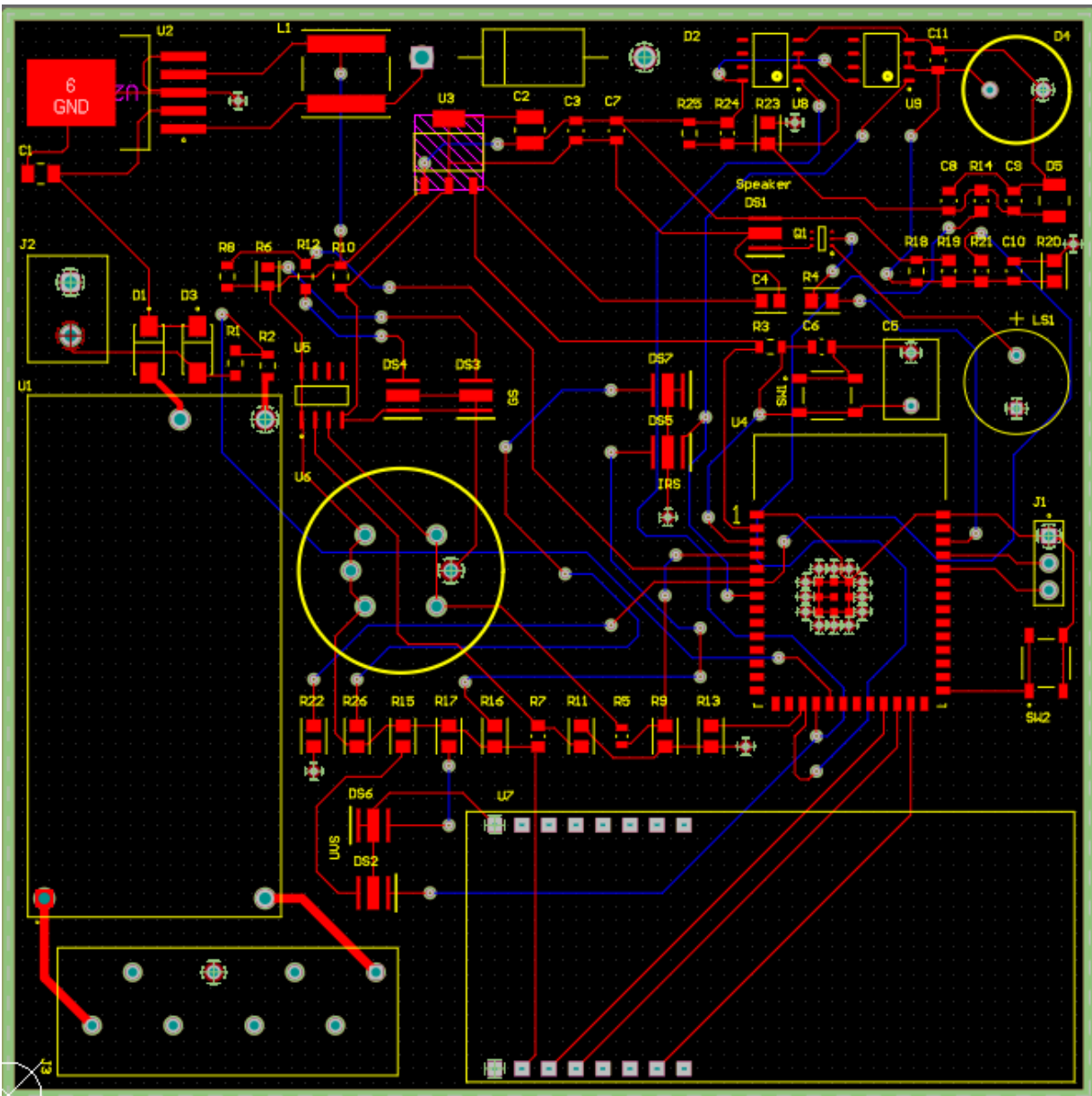


Figure 3: PCB Layout of the Hardware Subsystem

Validation

The validation of the hardware subsystem was conducted in both the design and post-fabrication stages of the process to make sure the designs worked as intended. Initial validation first began by testing each of the three custom designed sensor modules by simulating them in Multisim as seen in the figures below.

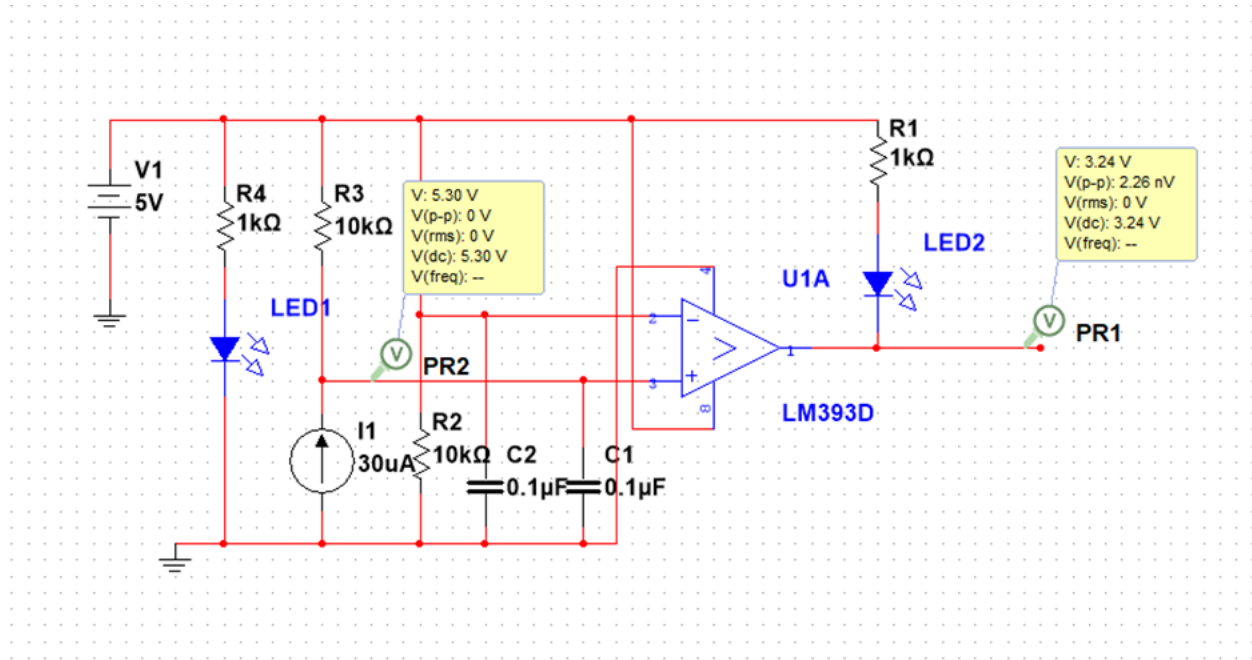


Figure 4: Infrared Sensor Module Simulation

First, the infrared sensor module was tested. This involved reading the voltage at the positive input terminal of the comparator where the infrared sensor would be sending its signal, as well as the output terminal of the comparator. While reading the voltages at these locations, the circuit was first simulated with no signal being generated, which correctly resulted in a low output signal from the comparator. Then, the circuit was simulated with the sensor generating a signal, which correctly resulted in a high output signal from the comparator. Therefore, this sensor module design passed the test and was utilized in the schematic.

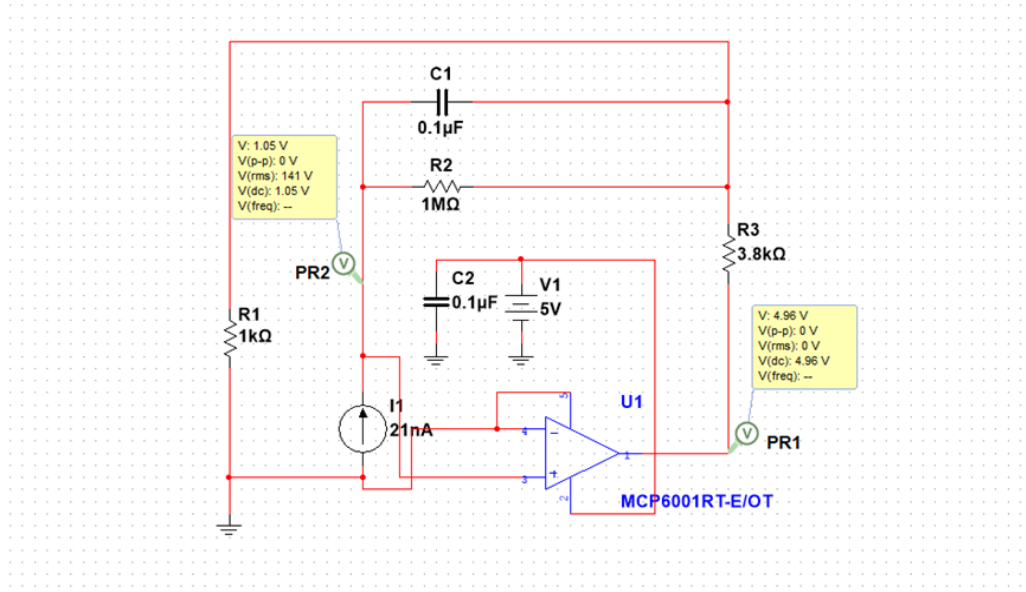


Figure 5: Ultraviolet Sensor Module Simulation

Next, the ultraviolet sensor module was tested. Similar to the infrared sensor module, this involved reading the voltage at the branch that the sensor would be sending its signal, as well as the output terminal of the comparator. While reading the voltages at these locations, the circuit was again first simulated with no signal being generated, which correctly resulted in a low output signal from the comparator. Then, the circuit was simulated with the sensor generating a signal, which correctly resulted in a high output signal from the comparator. Therefore, this sensor module design also passed the test and was utilized in the schematic.

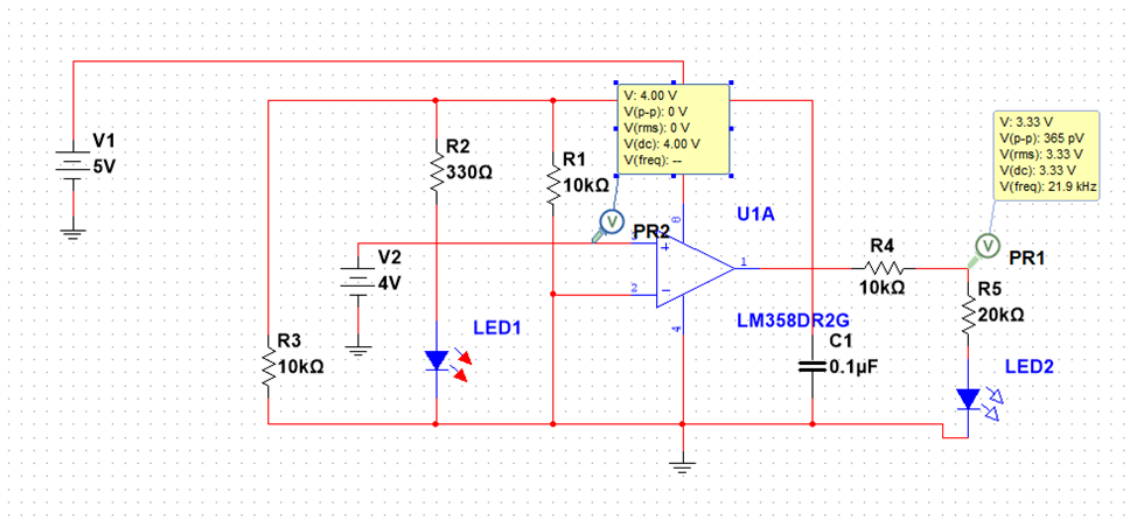


Figure 6: Gas Sensor Module Simulation

Finally, the gas sensor module was tested. This once again involved reading the voltage at the positive input terminal of the comparator, as well as the output terminal of the comparator. While reading the voltages at these locations, the circuit was simulated with no signal being generated, which correctly resulted in a low output signal from the comparator. Then, the circuit was simulated with the sensor generating a signal, which correctly resulted in a high output signal from the comparator. As a result, this sensor module design passed the test as well and was utilized in the schematic. Below is a table listing the module power voltages, raw sensor signal outputs, and comparator high signal outputs.

Table 1: Sensor Module Simulation Results

Sensor Type	Infrared	Ultraviolet	Gas
Power Voltage	5V	5V	5V
Sensor Raw Output	30uA	21nA	4V
High Signal Output	3.24V	4.96V	3.33V

Once the design was completed and the PCB was soldered, the PCB was then tested and validated. Some initial tests and careful reanalysis of both the PCB and circuit schematic revealed that some mistakes were made in the design, particularly regarding the placement of some resistors in the gas sensor and infrared sensor modules. These resistors were meant to reduce the input power voltages coming into the modules, lowering them to appropriate reference voltages for the corresponding comparators. Due to the misplacements in the schematic, this resulted in the reference voltages receiving no reduction, making it impossible for the comparators to output a high signal. These incorrect connections were fixed using bluewiring, allowing the circuit to work as intended. The successfully soldered and bluewired PCB is shown below.

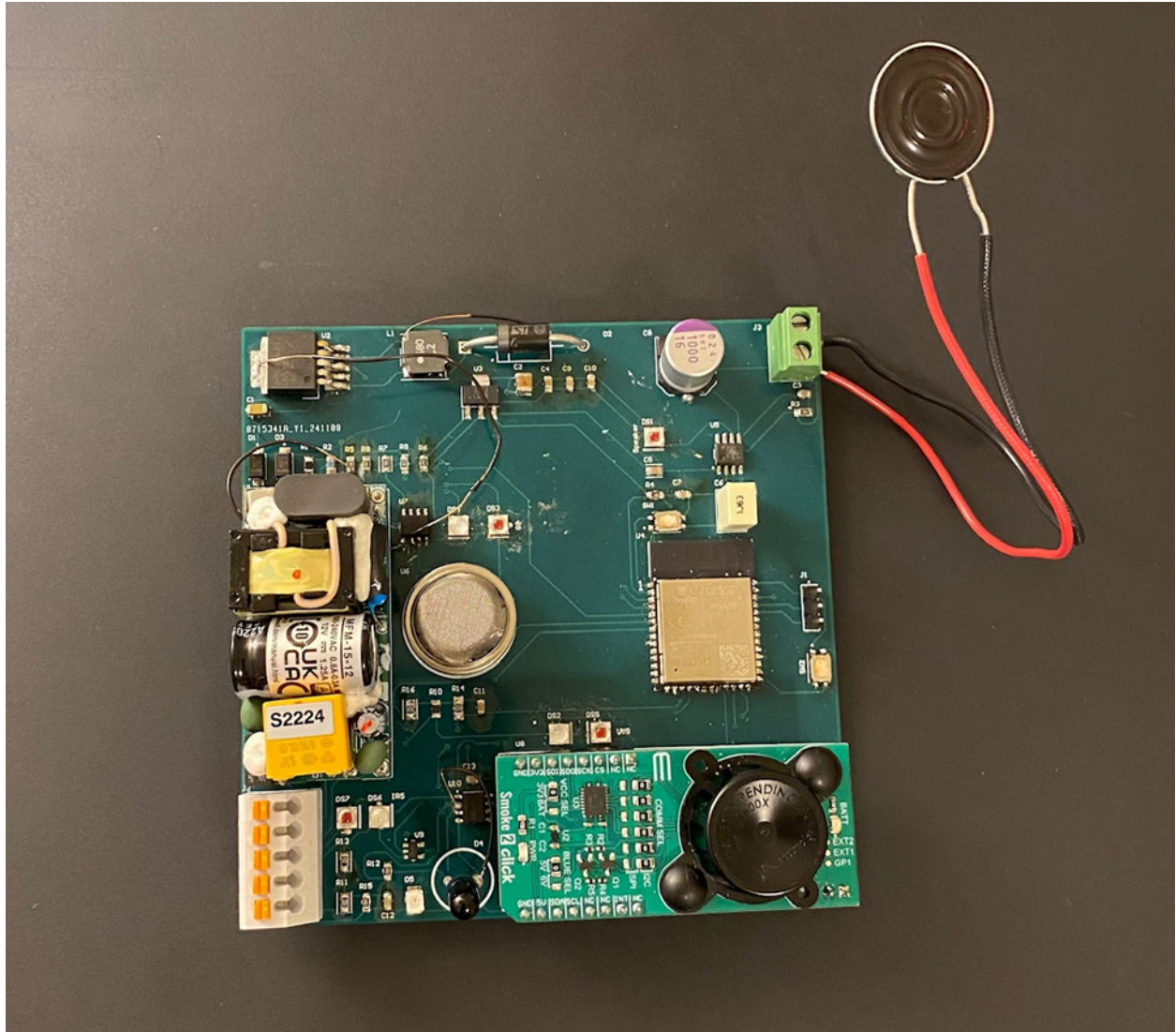


Figure 7: Completed PCB

Once all incorrect connections were fixed and tested with a multimeter to make sure nothing was incorrectly soldered, the true testing and validating could begin. First, the power distribution of the circuit was tested. To make validation easier, LEDs were utilized in the circuit design to clearly show if each sensor module was being successfully powered without having to use a voltmeter. During the test, 9V generated from a power supply was connected to the circuit, simulating the 9V backup battery. The PCB passed the test as all of the red LED's indicating power to the circuit sections turned on as seen below.

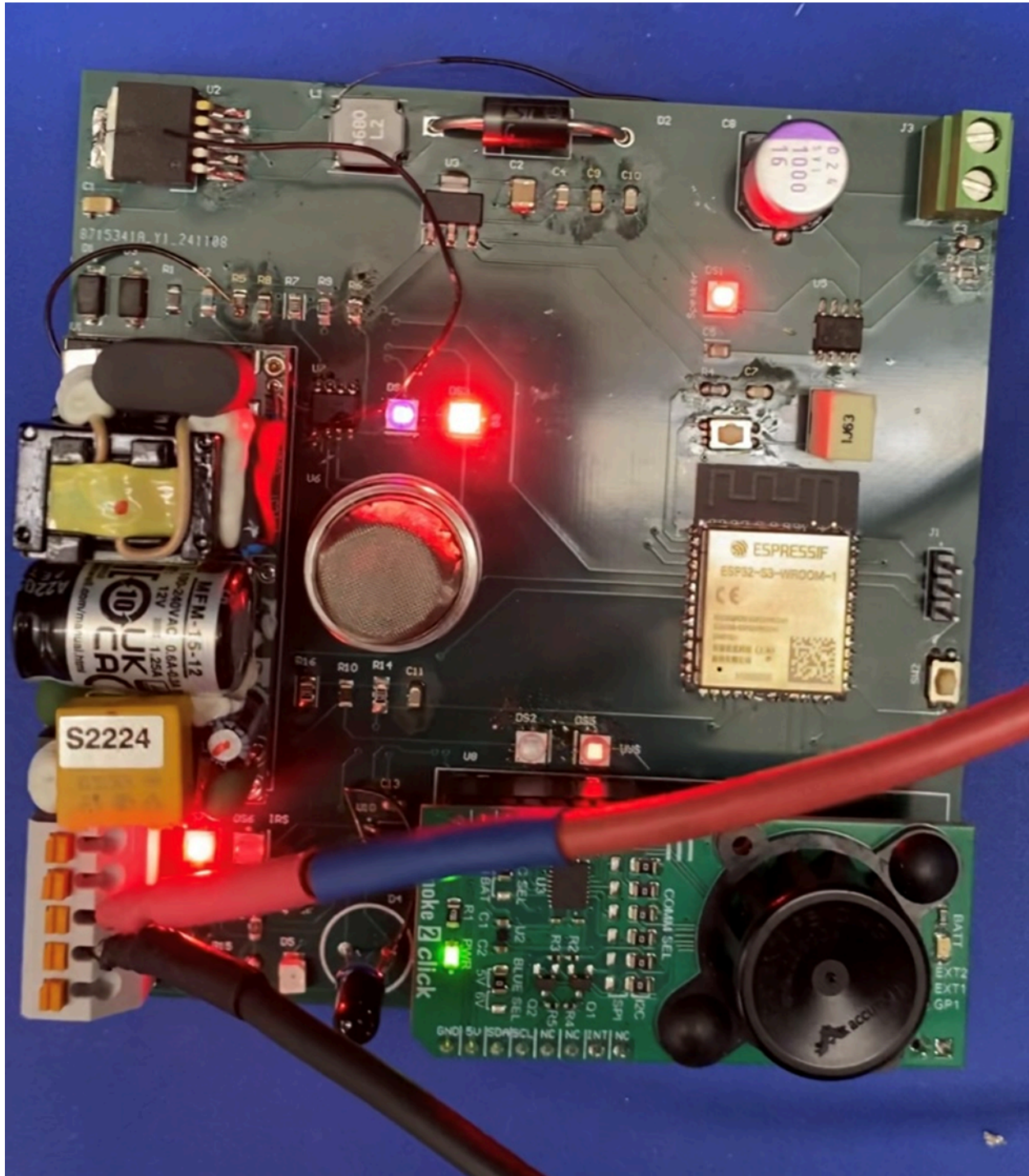


Figure 8: PCB successful power distribution test (Red LEDs on)

Eventually, to confirm validation, a multimeter was used to check the voltage delivered to each part of the circuit. In particular, the main power before and after the AC/DC converter, as well as

the voltage delivered to the microcontroller, photoelectric sensor, gas sensor, UV sensor, and IR sensor. All voltages read with the multimeter were as expected, as seen below.

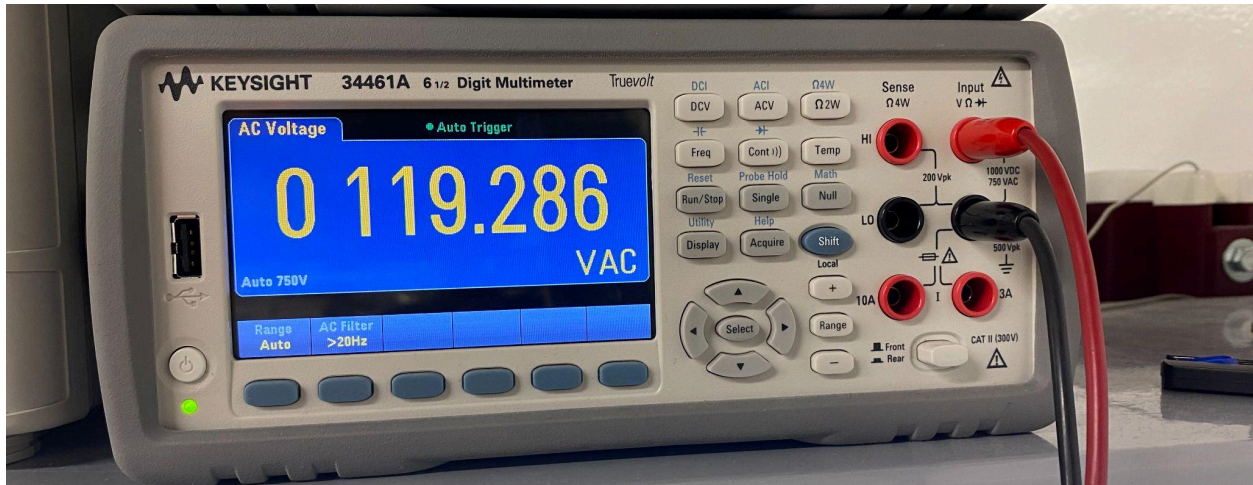


Figure 9: 120VAC successfully delivered to circuit



Figure 10: 12VDC successfully delivered from AC/DC converter

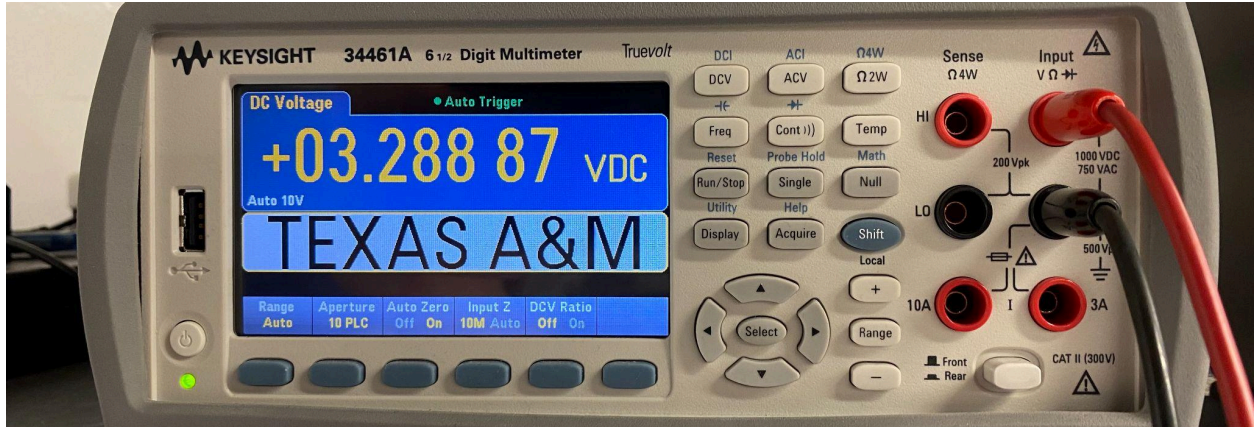


Figure 11: 3.3V successfully delivered to MCU

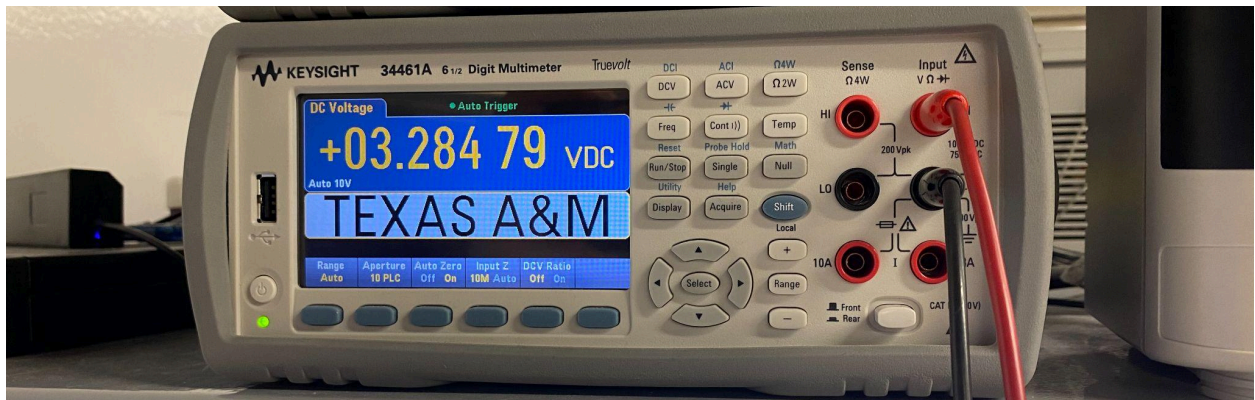


Figure 12: 3.3V successfully delivered to photoelectric sensor



Figure 13: 5V successfully delivered to IR sensor

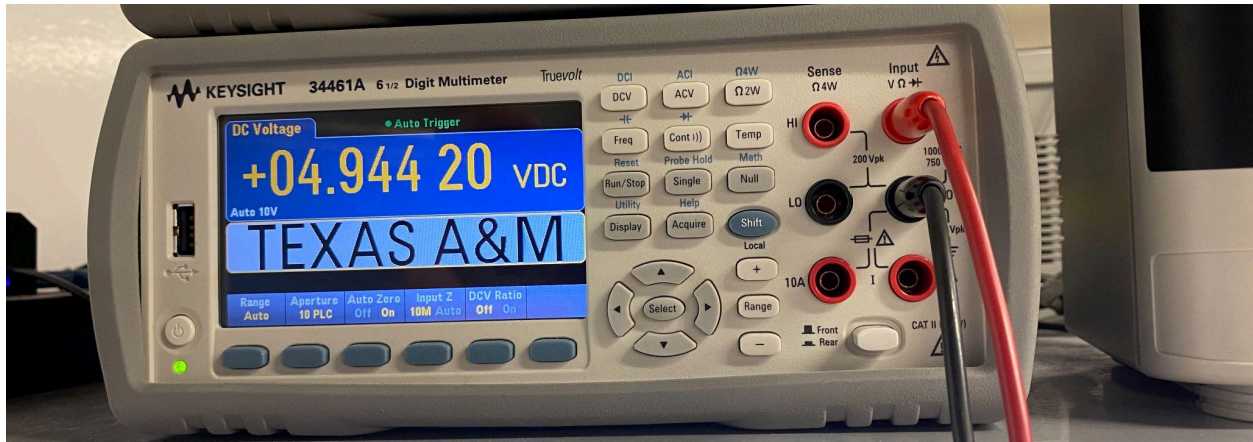


Figure 14: 5V successfully delivered to UV sensor

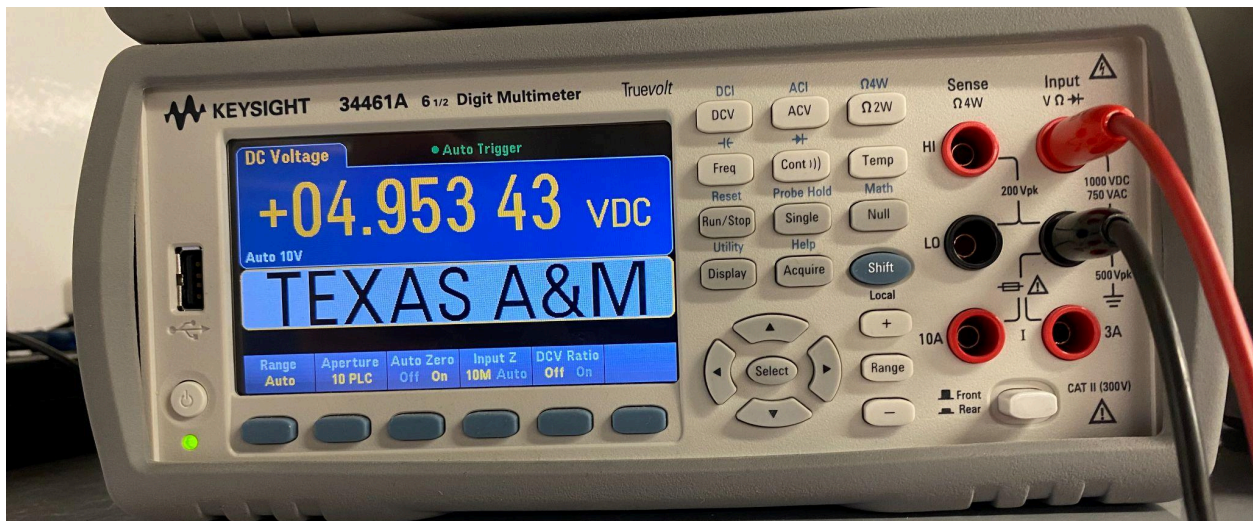


Figure 15: 5V successfully delivered to gas sensor

Next, each of the sensors were tested. Similar to the power distribution validation, LED's were utilized in the circuit design to clearly show if each sensor module was successfully outputting a signal when a sensor would output a raw signal. Once again, during each test, 9V generated from a power supply was connected to the circuit, simulating the 9V backup battery. Additionally, the power supply would simulate a second voltage used as a simulated raw output from the sensor. The sensor modules passed the tests as the blue LED's indicating a high output signal from the sensor module turned on as seen below for the infrared sensor module.

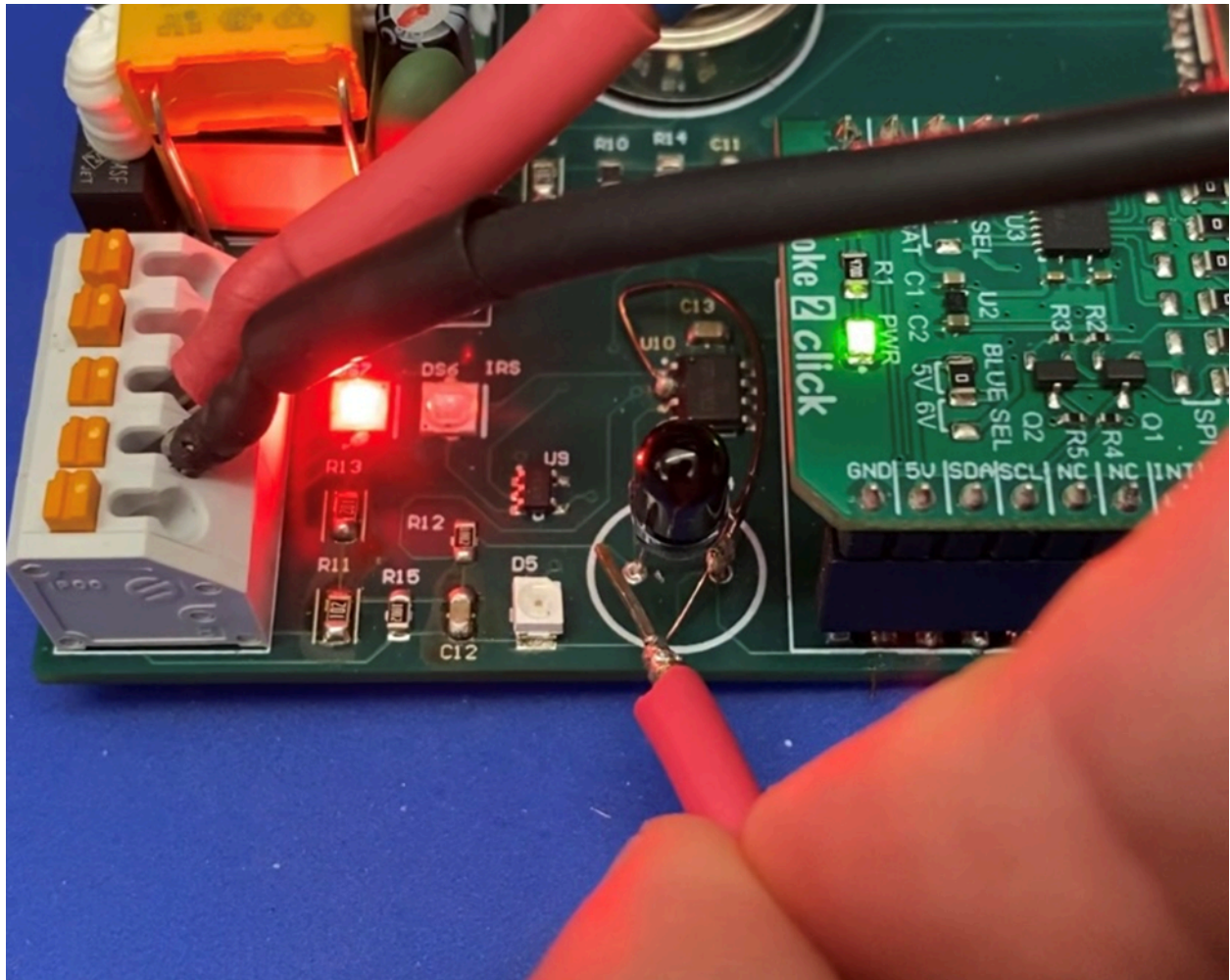


Figure 16: Infrared Sensor successful low output signal test (blue LED off)

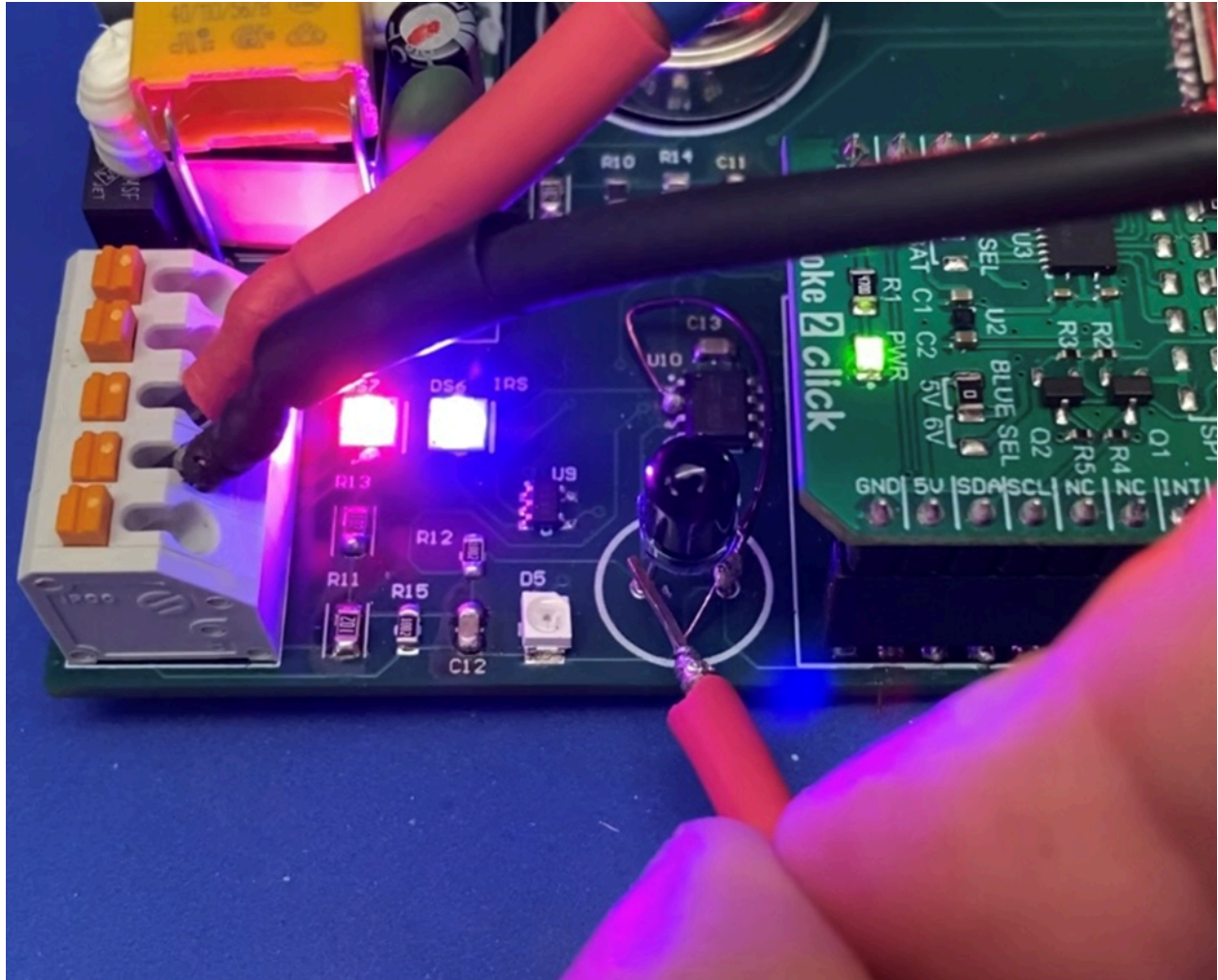


Figure 17: Infrared Sensor successful high output signal test (blue LED on)

Next, the alarm speaker was tested. Using the simulated 9V battery again and a simulated signal from the microcontroller using an oscilloscope wavegen output, the speaker was able to make an amplified sound, confirming that it worked as intended.

When designing the casing for the PCB, the dimensions were also taken into account from the validation plan. As a result, it was ensured that the diameter would not exceed 6.5 inches and the height would not exceed 3 inches.

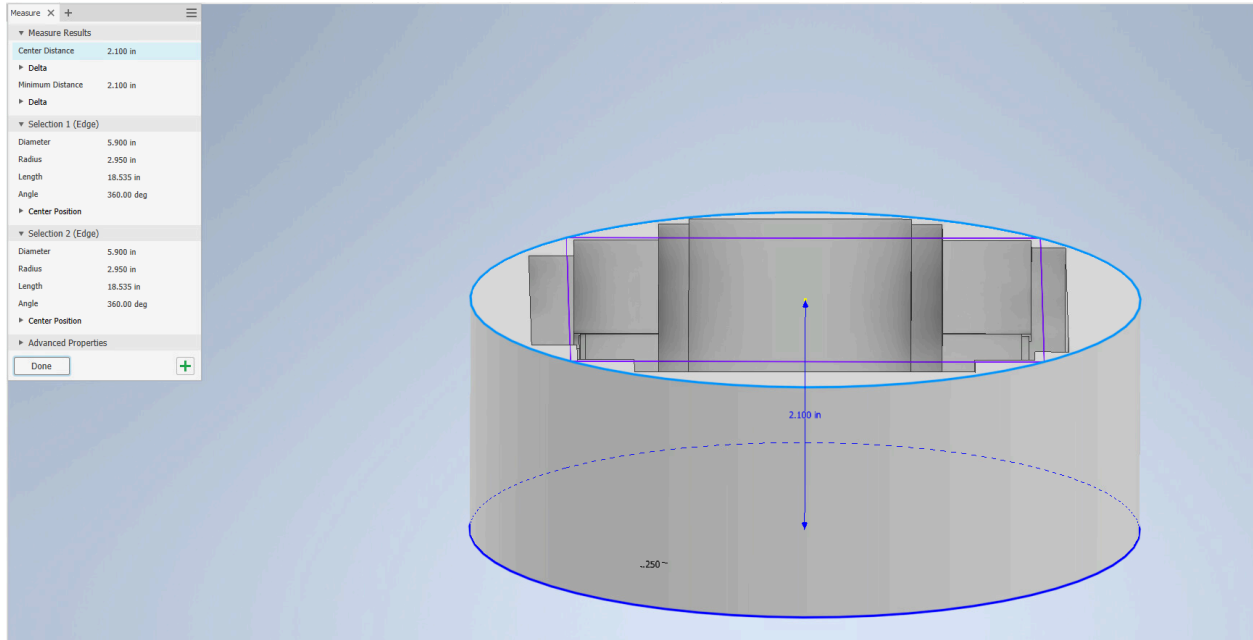


Figure 18: Main casing height reaches 2.1 inches

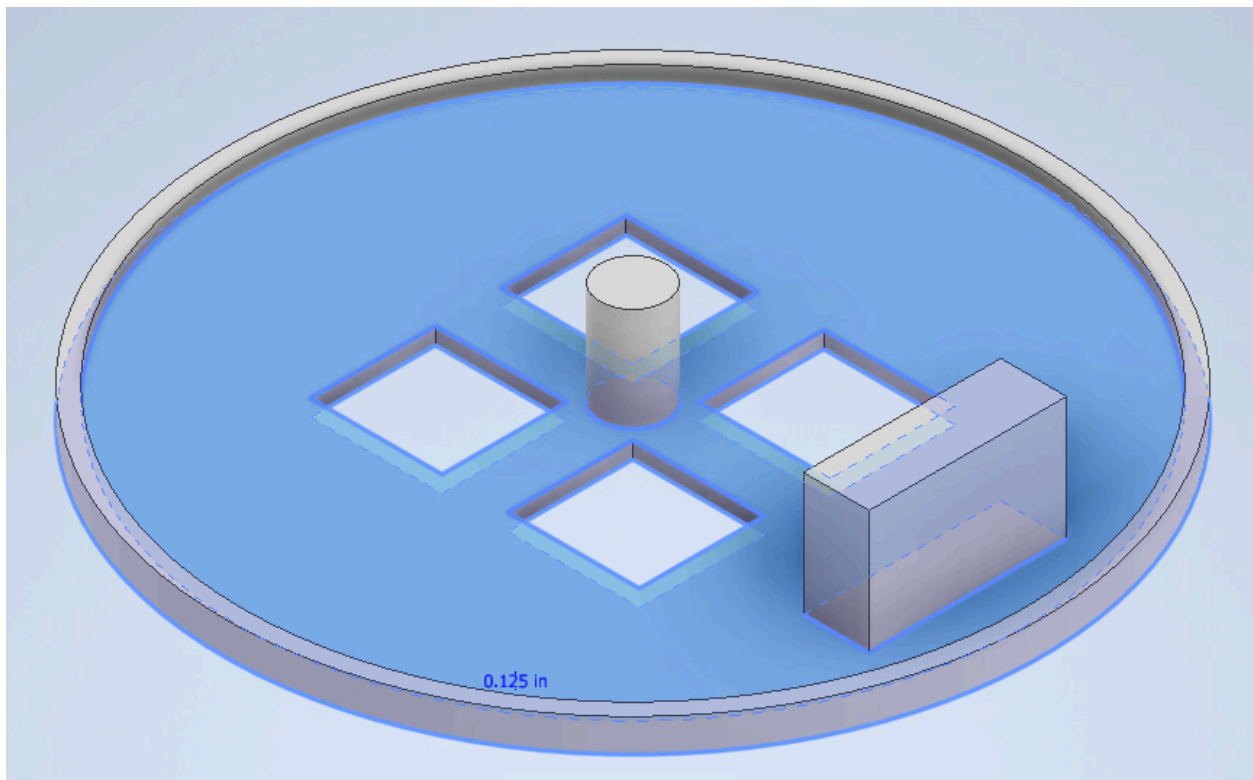


Figure 19: Casing Lid height reaches 0.125 inches for a total of 2.215 inches for the whole case

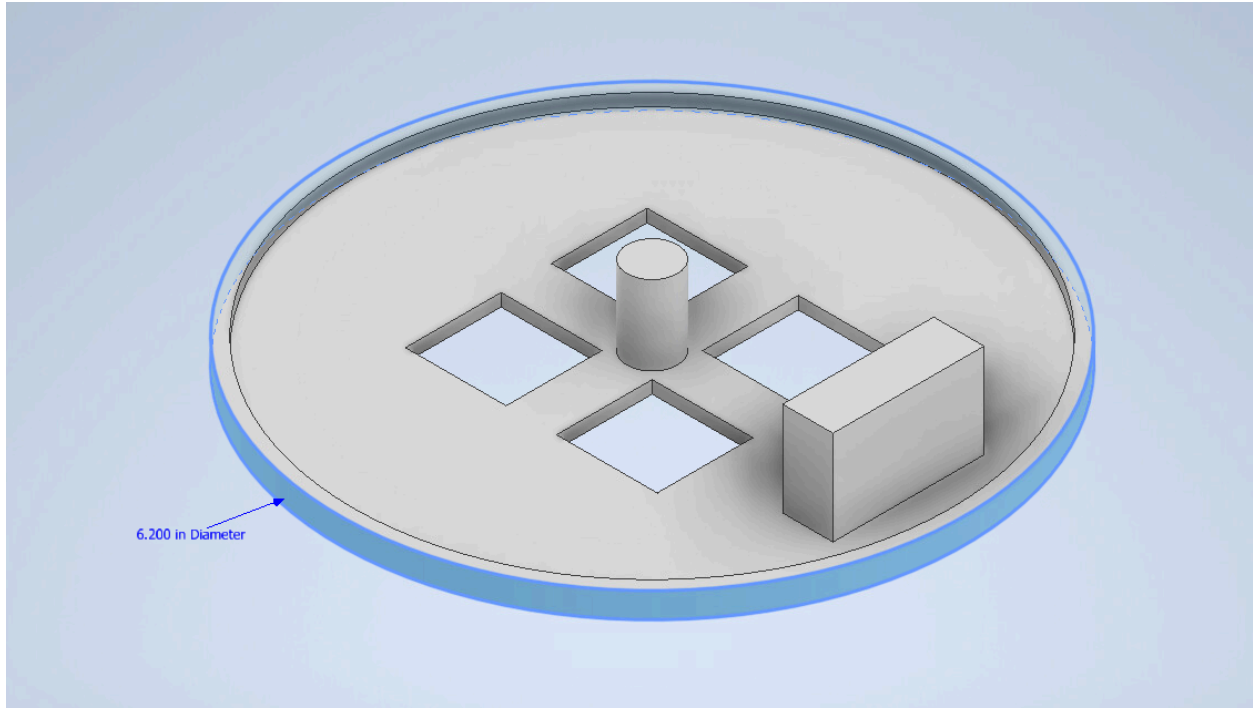


Figure 20: Casing Lid diameter reaches 6.2 inches

Conclusion

While a couple of the sensor modules have some minor issues and inconsistencies with their outputs, the PCB overall works as intended, given that the PCB successfully receives power in all locations, most sensor modules work as intended, and the speaker module works correctly. It will only need some minor modifications before it is interfaced with the rest of the Pyro Sensor system and will fulfill the fire sensing portion of the project.

PYRO SENSOR
OSCAR HERNANDEZ

MICROCONTROLLER FIRMWARE

Subsystem Introduction

The Microcontroller Software subsystem facilitates the integration between the hardware subsystem and the mobile application subsystem. It is a critical part of the fire detection system, responsible for continuously monitoring sensor data determining device status, and transmitting results to a central router.

Subsystem Details

The subsystem is implemented using ESP32-S3 microcontrollers, chosen for their built-in Wi-Fi capabilities and support for configurable pins. Each MCU is responsible for monitoring four sensors, including a smoke sensor, gas sensor, infrared sensor, and ultraviolet sensor, and activates the speaker upon detection of a fire event. The firmware also tracks the voltage level of the backup battery.

The firmware prioritizes the analog outputs of the gas, IR and UV sensors for logic processing; the digital outputs are available for logic based on hardware. The ESP32-S3, contains two ADC units capable of reading up to 3.1V, therefore I have assigned analog sensor readings to ADC1 and backup battery reading to ADC2 to minimize delay. ADC1 involves a task that continuously reads the analog voltages of the gas, IR, and UV sensors. ADC2's task is assigned to check the voltage of the backup battery continuously. Additionally, the smoke sensor's I2C SDA and SCL lines connect to non-strapping pins of the MCU. The MCU also biases a transistor that controls the active buzzer after a fire is detected.

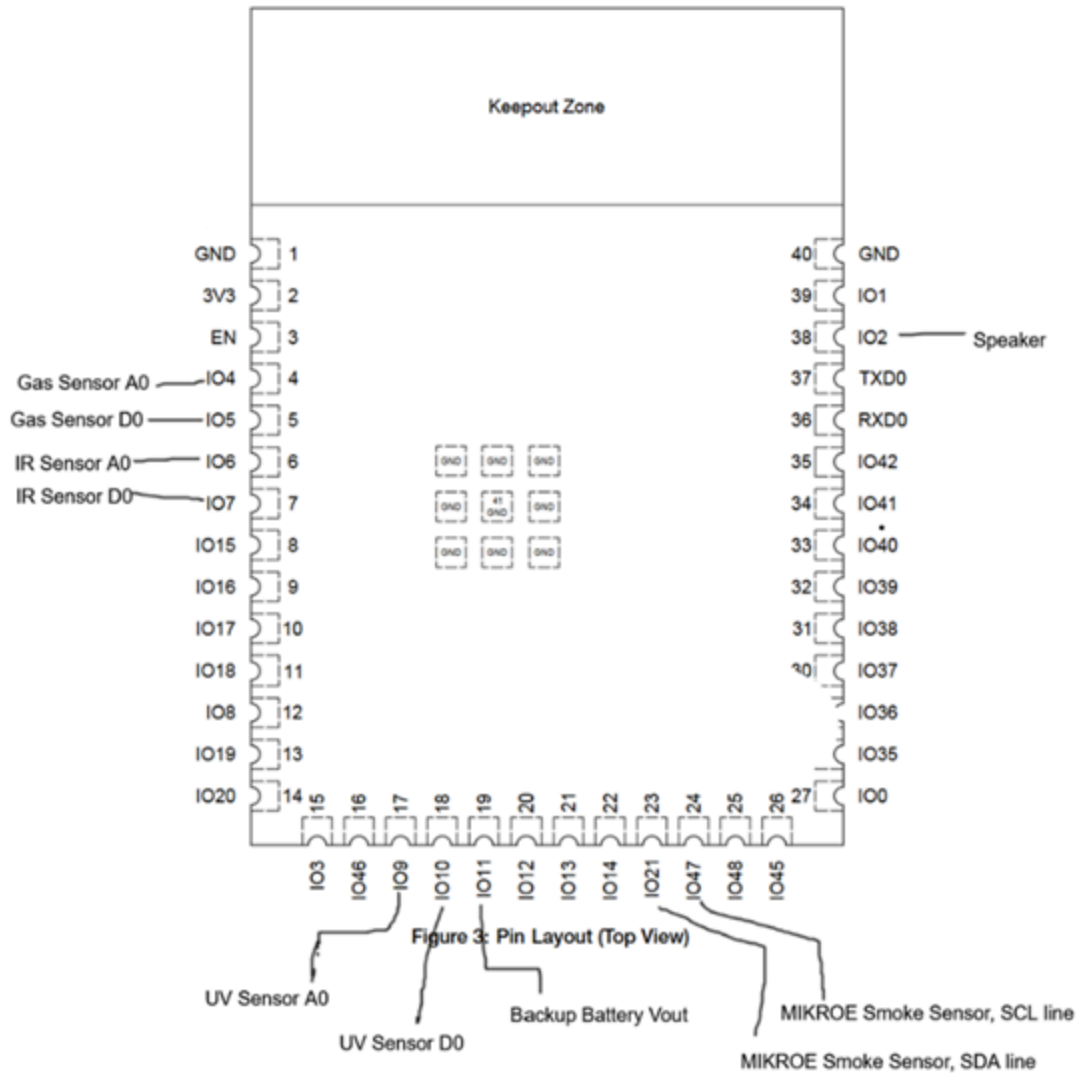


Figure 1: ESP32-S3 Physical pins

Subsystem Validation

The state machine for fire detection at every node is shown in Figure 2. Detector first-time setup, or hard reset is done prior to entering the state machine: i.e., receiving Wi-Fi credentials, connecting to Firebase, setting up GPIO pins, and describing macros. At the beginning of the loop, the speaker is turned off, and the sensors are calibrated under standby mode (i.e., room setting in the absence of fire) to get base values.

The fire detector operates in three main states based on an "Alarm Certainty" value calculated from sensor data. When the certainty is 75% or higher, the system enters an Alert state, turning the buzzer ON and sending an alert to Firebase. If the certainty is between 50% and 75% for at least 5 seconds, it enters a Warning state and sends a

warning to Firebase without triggering the buzzer. When the certainty drops to 25%, the system transitions to a Safe state, turning the buzzer OFF and notifying Firebase that conditions are safe. This threshold-based approach enables the detector to assess and respond to potential fire threats with increasing urgency.

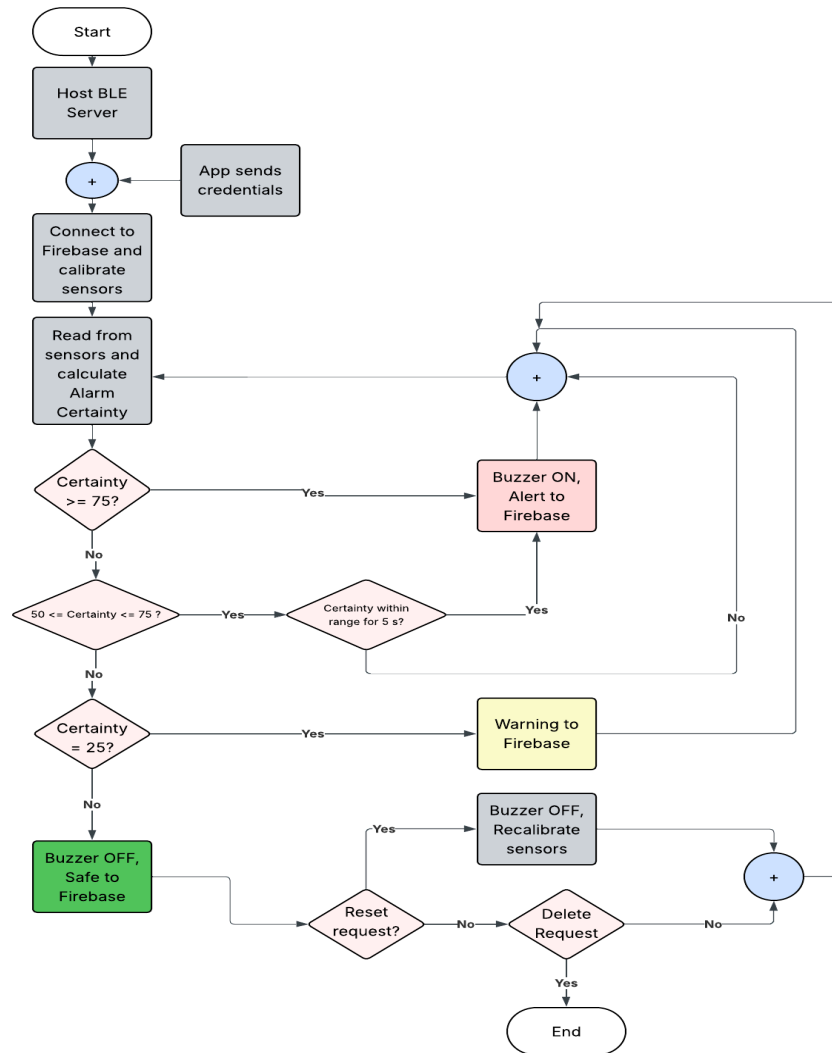


Figure 2: ESP32-S3 Physical pins

The firmware was scaled up to three nodes, with each node sending real-time data to Firebase as seen in Figure 3. Each node is labeled by its MAC ADDRESS, specific to each ESP32-S3. Thus, a room can be identified by giving an appropriate name by the app subsystem. Furthermore, each node shows the status and values of all peripherals.

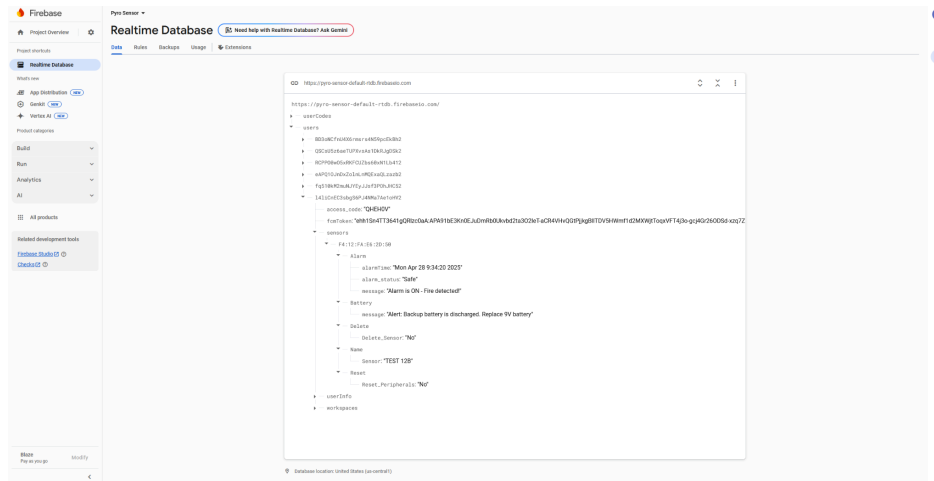


Figure 3: Firebase console

Subsystem Conclusion

The Microcontroller Firmware subsystem successfully integrates sensor monitoring, alarm activation, and network communication. It provides reliable and accurate fire detection while maintaining system resilience. The subsystem's design and implementation ensure it supports the larger fire detection system's objectives and lays a solid foundation for future enhancements.

The code performance is summarized in the below table:

Requirement	Specification	Result	Status
Device Setup Time	< 2 s	6.59 s avg.	Delayed
Firebase Push Latency	< 1 s	0.578 s avg.	Pass
Battery Monitoring	Good if > 7.2V	Brownout $\approx$ 6.7 V	Pass - Warning sent at 7.6 V
Alarm Confirmation Logic	$\geq 75\%$ to trigger	$\approx 99\%$	Pass
Time to Activate Buzzer upon Detection	< 5 s	< 1 us	Pass
Time to Alert App upon Detection	< 5 s	≈ 1 s	Pass

PYRO SENSOR
ALFONSO WONG

MOBILE APPLICATION

REVISION – Final Report

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1. Introduction

The Mobile Application Subsystem is a core component of the Fire/Smoke Detection and Alert System, which provides users with real-time monitoring and alert functionality. Developed in Kotlin, the mobile app ensures user interaction with the system, focusing on reliable communication with the sensors. The app features a user account system, which allows users to create accounts, log in, and set up workspaces. In each workspace, users can add, view, and manage sensors. They are displayed using a RecyclerView with CardView for a concise presentation of sensor data. Each card shows key information, including sensor status, connectivity, and alerts related to fire or smoke detection. When a fire or smoke event is detected, the app immediately notifies users by providing details like the specific room affected and the duration of the incident. This enables users to quickly assess the situation and take necessary action. A notification is also sent to the fire department with information such as the address, incident duration, type of building, and expected occupants.

2. System Architecture

2.1 Subsystem Overview

The purpose of the mobile application is to let users manage sensors, check their status, and receive updates when a fire or smoke is detected. The application connects to the backend system to get sensor information and alert data, which keeps users informed about any incident detected. In addition to notifying users, the application also notifies information to firefighters, including the type of building, address, and details about expected occupants, such as their names and ages. This ensures that emergency responders have the information they need to act quickly and effectively.

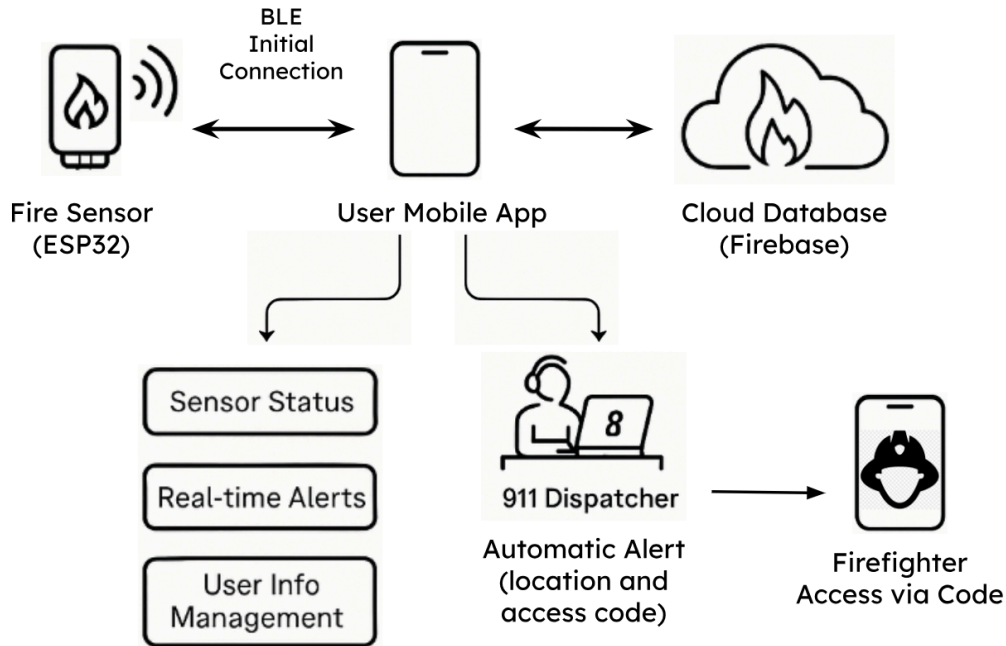


Figure 1. Mobile Application Subsystem

3. Design and Development

3.1 Programming Language and Tools

The mobile application is developed using Kotlin as the programming language. Android Studio is the IDE used for coding, debugging, and interface design. XML is used to create the layout of the app, and components from XML such as RecyclerView and CardView are used to display sensor information. Firebase is integrated to provide user authentication, notification alerts, and real-time database communication.

```

<? class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState) //initializes activity and sets saved state
        setContentView(R.layout.activity_main) //sets layout for the main activity

        loadFragment(SensorsFragment()) //loads the default fragment (sensors tab)

        val bottomNavigation: BottomNavigationView = findViewById(R.id.bottom_navigation) //finds the bottom navigation view
        bottomNavigation.setOnItemSelectedListener { menuItem -> //handles item selection in navigation
            when (menuItem.itemId) { //checks the selected item id
                R.id.nav_sensors -> {
                    loadFragment(SensorsFragment()) //loads the sensors fragment
                    true
                }
                R.id.nav_profile -> {
                    loadFragment(ProfileFragment()) //loads the profile fragment
                    true
                }
                R.id.nav_settings -> {
                    loadFragment(SettingsFragment()) //loads the settings fragment
                    true
                }
                else -> false //returns false for unhandled cases
            }
        }

        private fun loadFragment(fragment: Fragment) { //replaces the current fragment with the specified one
            supportFragmentManager.beginTransaction()
                .replace(R.id.fragment_container, fragment)
                .commit()
        }
    }
}

```

Figure 2. Example Kotlin Code

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      xmlns:android="http://schemas.android.com/apk/res/android"
4      xmlns:app="http://schemas.android.com/apk/res-auto"
5      xmlns:tools="http://schemas.android.com/tools"
6      android:layout_width="match_parent"
7      android:layout_height="match_parent">
8      <!-- sets up email login and hint -->
9      <EditText
10         android:id="@+id/loginEmail"
11         android:layout_width="wrap_content"
12         android:layout_height="wrap_content"
13         android:ems="10"
14         android:hint="Email Address"
15         android:inputType="textEmailAddress"
16         app:layout_constraintTop_toBottomOf="@id/loginView"
17         app:layout_constraintStart_toStartOf="parent"
18         app:layout_constraintEnd_toEndOf="parent"
19         android:layout_marginTop="100dp"
20     />
21     <!-- sets up password login and hint -->
22     <EditText
23         android:id="@+id/loginPassword"
24         android:layout_width="wrap_content"
25         android:layout_height="wrap_content"
26         android:hint="Password"
27         android:ems="10"
28         android:inputType="textPassword"
29         app:layout_constraintTop_toBottomOf="@id/loginEmail"
30         app:layout_constraintStart_toStartOf="parent"
31         app:layout_constraintEnd_toEndOf="parent" />
32     <!-- sets up page title-->
33     <TextView
34         android:id="@+id/loginView"
35         android:layout_width="wrap_content"
36         android:layout_height="wrap_content"
37         android:text="Sign in"
38         android:textSize="30sp"
39         app:layout_constraintTop_toTopOf="parent"

```

Figure 3. Example XML code

3.2 App Structure

The application is organized into key components to manage its functionality. It includes activities for login, signup, workspace, and managing the main page. The login and signup activities handle account creation and secure authentication. The main activity is the central dashboard, displaying sensor details and alert notifications. A RecyclerView is used within the dashboard to present a list of sensors, while CardView is used to display each sensor's details. Adapter classes binding data to these views. Workspace activity allows you to name your building, select the type of building (ex. House or Apartment), and give information on expected occupants.

3.3 User Interface Design

The user interface is designed to make the application easy to use. The login and signup screens are very straightforward, requiring little input to create or access an account. The dashboard is structured to display sensor information, with each sensor's details, including its location, connectivity, and status, shown on separate cards. Alerts are presented efficiently to ensure users can quickly respond to critical events.

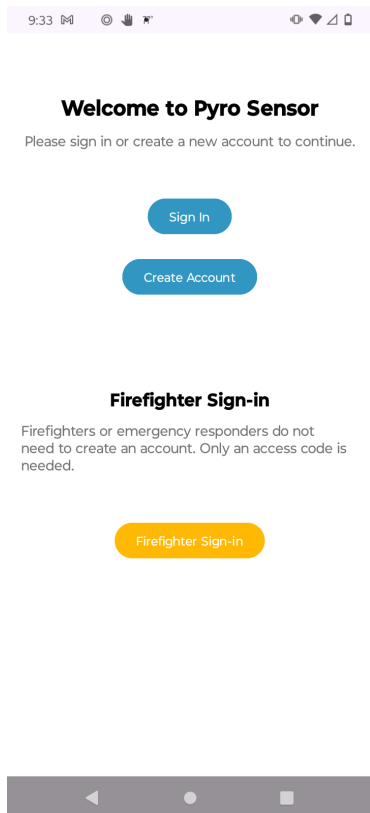


Figure 4. Welcome Screen UI

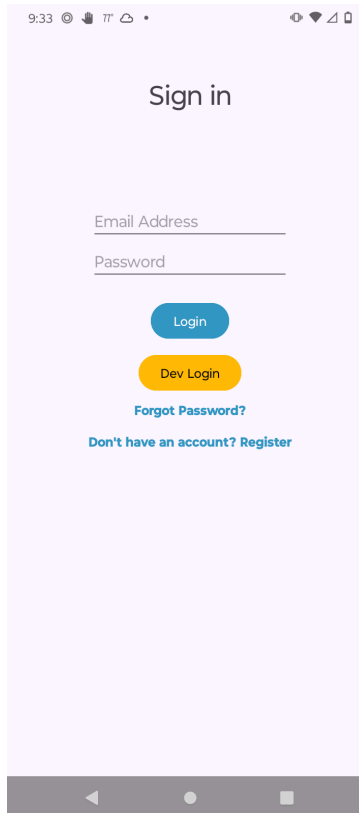


Figure 5. User Sign in UI

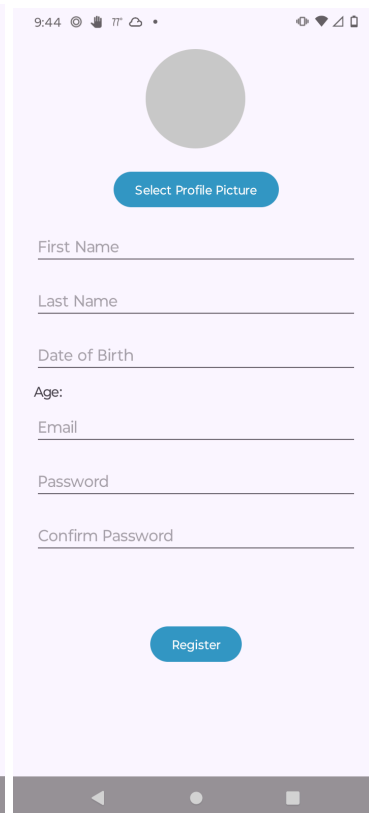


Figure 6. User Registration UI

3.4 App Features

The application includes features to support user account management, sensor monitoring, and real-time alerts. Users can create accounts, log in securely, and manage their profiles. The app allows users to monitor the status of sensors within their workspaces. Real-time alerts notify users of any detected fire or smoke, displaying details about the location and time of incident. Additionally, the application is designed to notify firefighters with information such as the building type, address, and details about expected occupants, such as names and ages.

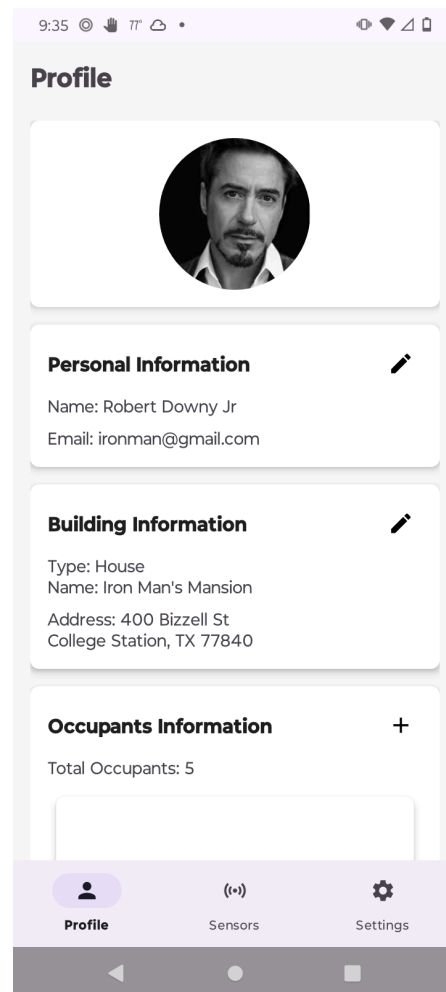


Figure 7. User Profile UI

4. Core Functionalities

4.1 User Authentication

The mobile application includes secure user authentication, allowing users to create accounts, log in, and access their data. Firebase Authentication is used to handle account

creation and login validation. As well as the storage of that information. This ensures that only authorized users can access the account, maintaining the integrity of sensor data and alert information.

4.2 Workspace and Sensor Management

The application enables users to create workspaces that store critical information about the building they are installing the sensors in. Each workspace contains details such as the building type, address, and information about expected occupants, including their names and ages. This includes any pets they may have. Within the main dashboard, users can manage the sensors associated with each workspace. Sensors are displayed dynamically using a RecyclerView, with each sensor's details shown on a CardView. Users can add new sensors to a workspace, delete sensors, and rename existing sensors to reflect their locations(ex. Kitchen Sensor). The dashboard also allows users to check the status of each sensor, including its operational status and connectivity.

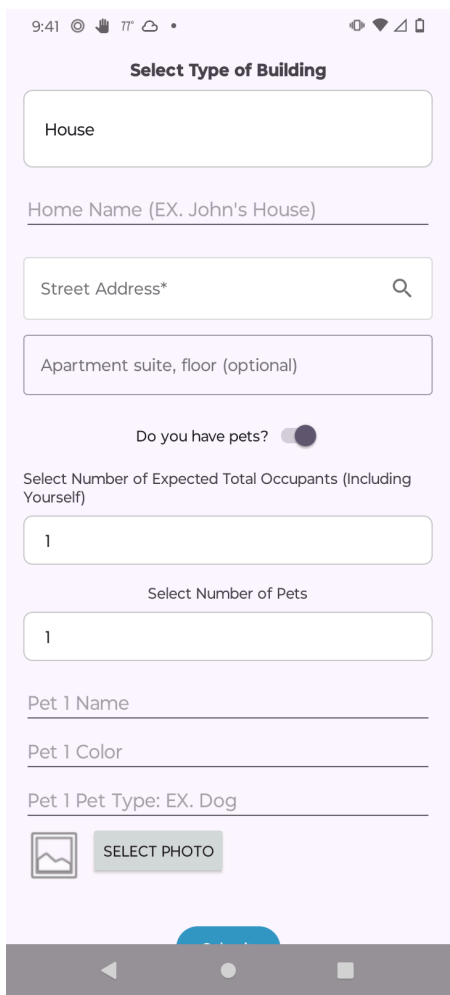


Figure 8. Workspace Setup UI

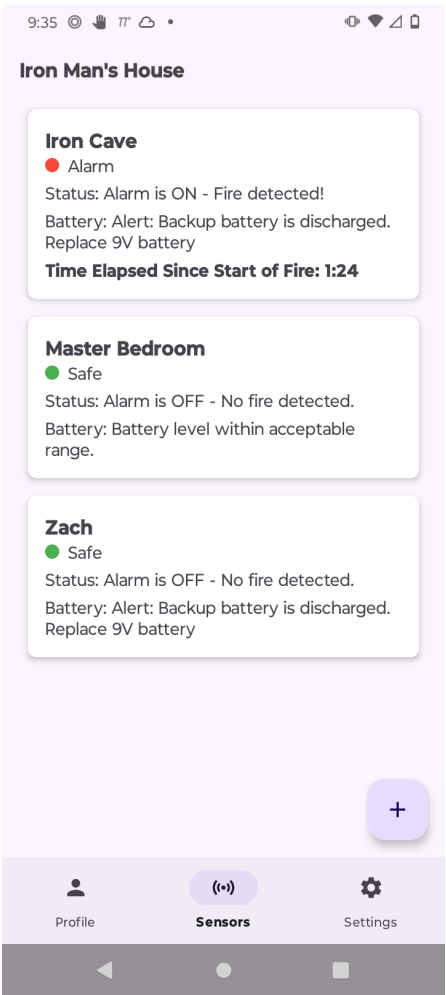


Figure 9. Sensor Display UI

4.3 Real-Time Alerts

When the system detects a fire or smoke, the app receives real-time alerts and are displayed on the dashboard. The alerts provide key details about the incident, such as the location of the sensor and the duration of the detection. This ensures that users are informed and can quickly respond to the situation. The app also communicates this information to firefighters, including details such as the building type, address, and expected occupants, so they can prepare for the emergency appropriately.

4.4 Connectivity and Data Communication

The application communicates with the backend through Firebase, using APIs to retrieve sensor data and send updates. This ensures that the app is always synchronized with the latest information from the system. The communication is designed to be reliable and efficient, providing real-time updates without the need for local data storage. If the connection is lost, the information is kept in the cloud and accessible at a later time.

5. Database and Data Management

5.1 Real-Time Communication

The application uses Firebase to manage and synchronize data in real time. This allows the app to receive updates about sensor status and fire/smoke alerts as they happen. Firebase is responsible for storing and retrieving data, ensuring that the application always displays the current information. In addition to data synchronization, Firebase also sends push notifications to users when an alert is triggered, informing them of any detected fire or smoke.



Figure 10. Firebase Data Structure

6. Challenges

6.1 Technical Challenges

Three of the key technical challenges was ensuring reliable real-time communication between the app and the backend system, especially for delivering alerts without delay. Another challenge was managing dynamic sensor data in the main dashboard while allowing users to add, delete, and rename sensors efficiently. Additionally, determining how to select the correct firefighter station to send alerts presents a challenge, as it requires integrating location data and ensuring that the appropriate responders would receive the relevant information based on the building's address.

7. Conclusion and Future Work

7.1 Summary

The mobile application subsystem plays a very important role in the Fire/Smoke Detection and Alert System, allowing users to manage sensors, receive real-time alerts, and provide critical information to firefighters. By using Firebase for real-time data and push notifications, the app ensures that users are kept informed of any incidents in their buildings. The application also supports efficient sensor management and effective workspaces, ensuring that users can easily monitor sensor status and make adjustments when necessary.

7.2 Future Enhancements

In the future, the app will incorporate additional features such as adding authorized users to view/manage sensors and buildings protected by our Pyro Sensor. We will also add the ability to manage multiple buildings with just one user. The most innovative enhancement, would be adding a cellular module on the sensor, to be able communicate with the database regardless of the collapse of an outer wifi service.

PYRO SENSOR
OSCAR HERNANDEZ
ALFONSO WONG
MARTIN FORD

SYSTEM DESCRIPTION AND DEVELOPMENT

REVISION 3

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1. Overview

The Pyro Sensor System is an innovative and reliable solution for detecting and responding to fire and smoke incidents in indoor environments. The system is composed of multiple interconnected subsystems: Hardware Sensing, Mobile Application, Bluetooth Communication, Cloud Database Integration, and Emergency Notification Services. Successful integration of each subsystem was critical to delivering a complete, functional product, while thorough validation and testing were essential to ensure the system met design requirements and performance standards. After completing both integration and validation, the Pyro Sensor System was finalized as a robust and user-friendly safety solution for residential and commercial properties.

2. Development Plan and Execution

2.1. Design Plan

The Pyro Sensor System was developed in response to a problem proposed to the team: creating a more effective method for detecting fires and alerting emergency responders. Initial research focused on existing fire detection technologies and their limitations, including delayed response times and lack of detailed information for first responders. The system design integrates multiple subsystems to address these gaps: a network of wireless fire and smoke sensors, Bluetooth communication for initial setup, a mobile application for user monitoring and control, and a cloud database for real-time data storage and retrieval. Moisture and temperature detection were considered but ultimately smoke and heat sensing were prioritized to maximize fire detection reliability. A key feature is automated emergency notification, where the system sends location and occupant information directly to 911 dispatchers. Firefighters are also able to retrieve additional building details using a secure access code linked to the database. This integrated approach ensures faster detection, streamlined communication, and improved situational awareness, offering a more complete solution than standard residential smoke detectors.

2.2. Execution

Before full system integration, individual subsystems were finalized. This included completing the mobile application, finalizing Bluetooth (BLE) communication for sensor setup, configuring Firebase for real-time data management, and implementing Twilio for emergency notifications. Subsystem improvements were made, such as optimizing sensor polling intervals to reduce power usage and improving Bluetooth reliability.

On the hardware side, the original PCB design had issues and required some redesigning, retesting, and validation. Additionally, the speaker initially planned for

alarm output was replaced with a buzzer, as the ESP32 microcontroller could not supply enough voltage to drive a speaker directly. With the new buzzer, the ESP32 would then send a control signal to a transistor acting as a switch to deliver power from the main circuit power to the buzzer for activation.

Integration then focused on connecting sensor data to Firebase for real-time updates, allowing the mobile app to monitor all sensors remotely. Twilio integration enabled automatic emergency alert messaging, completing the system’s emergency communication pathway.

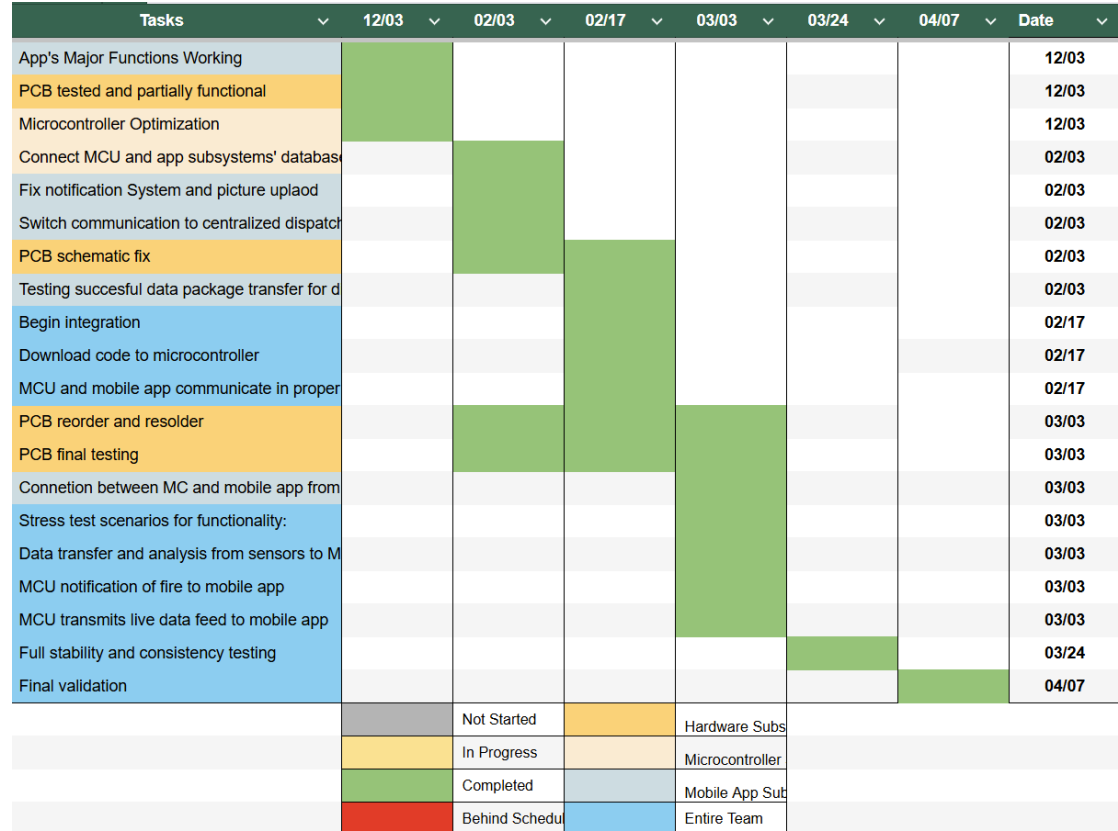


Figure 1: Schedule of Execution and Testing

2.3. Validation Plan

Test Name	Successful Criteria	Result
Sensor Functionality	Ionization and photoelectric sensors activate on smoke contact and output signals. IR and UV sensors detect flames up to 10 ft away and output signals.	Successful
Speaker Functionality	When triggered by MCU fire detection, speaker emits sound from signal at >65 dB.	Successful Max Volume: 93 dB
Operating Time	System operates and receives full power continuously for 1 hour on each power source.	Successful Main Power: >1 hour tested Backup Battery: >2 hours tested
Network	MCU is able to aggregate data from sensors to router.	Successful
Thermal Resistance	System operates successfully in environment within temperatures ranging from 40F to 120F.	Successful Approximate max tested temp: >120F
Trouble Signal	MCU sends trouble signal to app for malfunctions.	Successful
User Profile Creation	Users are able to create an account and be able to sign in.	Successful
Push Notifications	Be able to push notifications to users and send text to the central dispatcher when alarm is triggered by MCU.	Successful
MCU Peripherals Interface	MCU reads signals from all sensors, and activates/deactivates alarms.	Successful
Device and Application Communication	A new device can be added to the user profile in application via BLE.	Successful

Real Time Communication	MCU is feeding constant information to the database and it is being displayed in real time in mobile application.	Successful
Trigger Warning from MC	MCU must send a trigger alert of ongoing event and application must send notification to user and dispatch	Successful
Early Fire Detection	System detects smoke/flame with 95% accuracy. Detected alarm activates buzzer.	Successful Approximated Accuracy: >95%

3. Critical System Data

3.1. Speaker Functionality

Speaker Functionality	When triggered by MCU fire detection, speaker emits sound from signal at >65 dB.	Successful Max Volume: 93 dB
-----------------------	--	---------------------------------

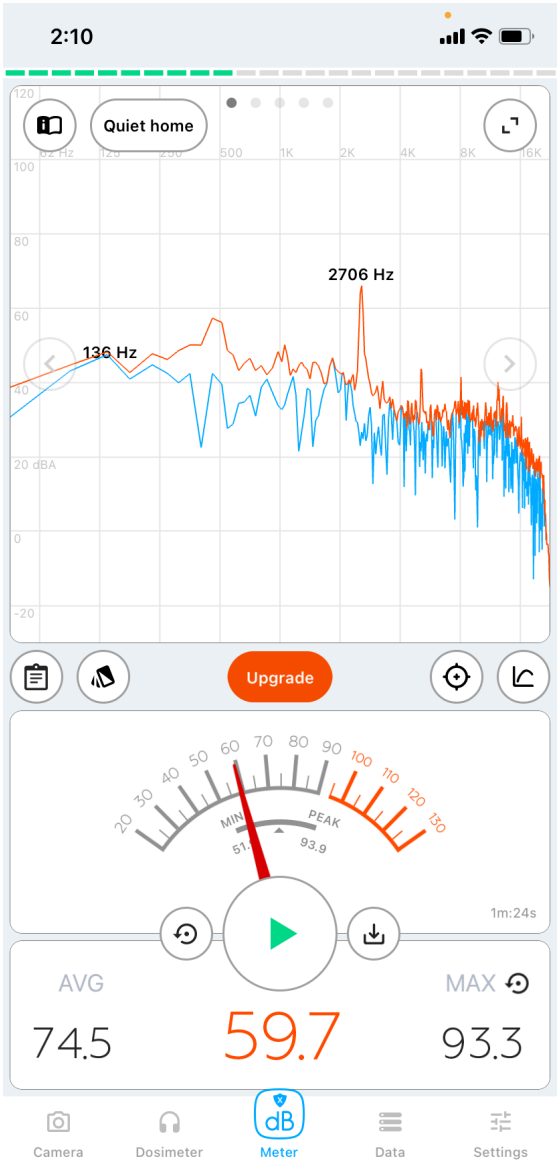


Figure 1. Max decibel count recorded: 93.3 dB

3.2. Operating Time

Operating Time	System operates and receives full power continuously for 1 hour on each power source.	Successful Main Power: >1 hour tested Backup Battery: >2 hours tested
----------------	---	---

Table 1. Recorded Operating Times For Backup Battery Variants

Battery Chemistry	Expected Run-Time on Battery Power (Hours)
Alkaline	0.7
NiMh	1
Lithium-ion	2.81
Lithium-polymer	2.87

3.3. MCU Peripherals Interface

MCU Peripherals Interface	MCU reads signals from all sensors, and activates/deactivates alarms.	Successful
---------------------------	---	------------

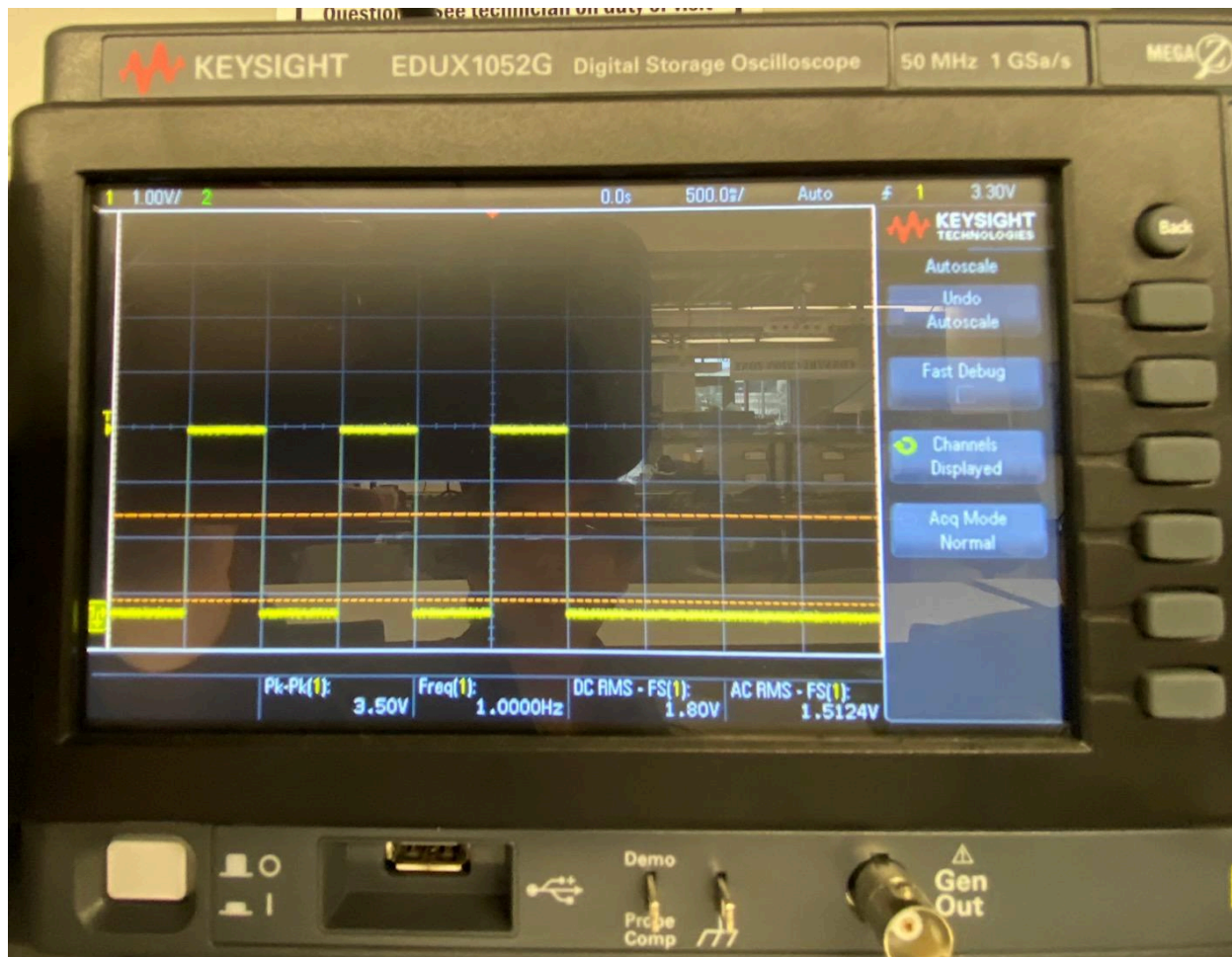


Figure 2. Oscilloscope recording of MCU control signal to transistor

Analog Value Display using ESP32

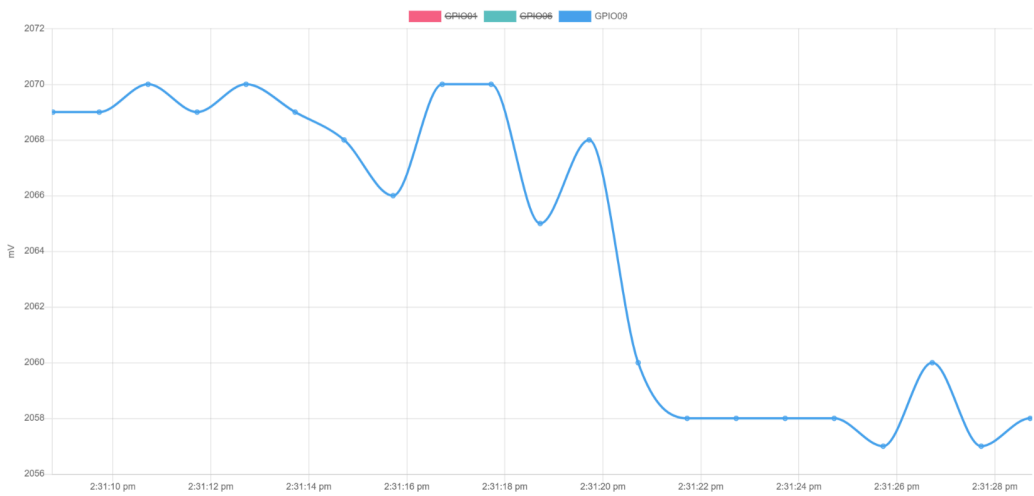


Figure 3. Battery power (test 9V dropped to 7V)

Analog Value Display using ESP32

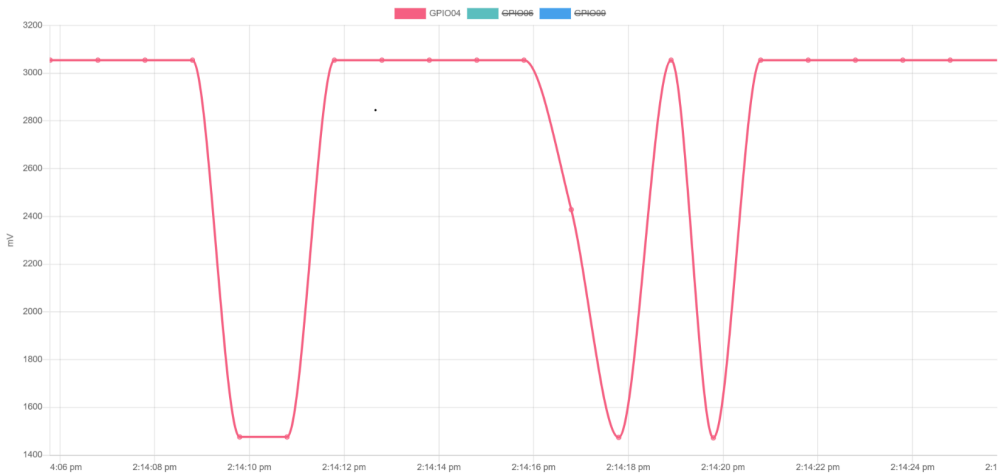


Figure 4. Gas Sensor readings

Analog Value Display using ESP32



Figure 5. IR and UV Sensor readings

3.4. Fire Detection Reliability

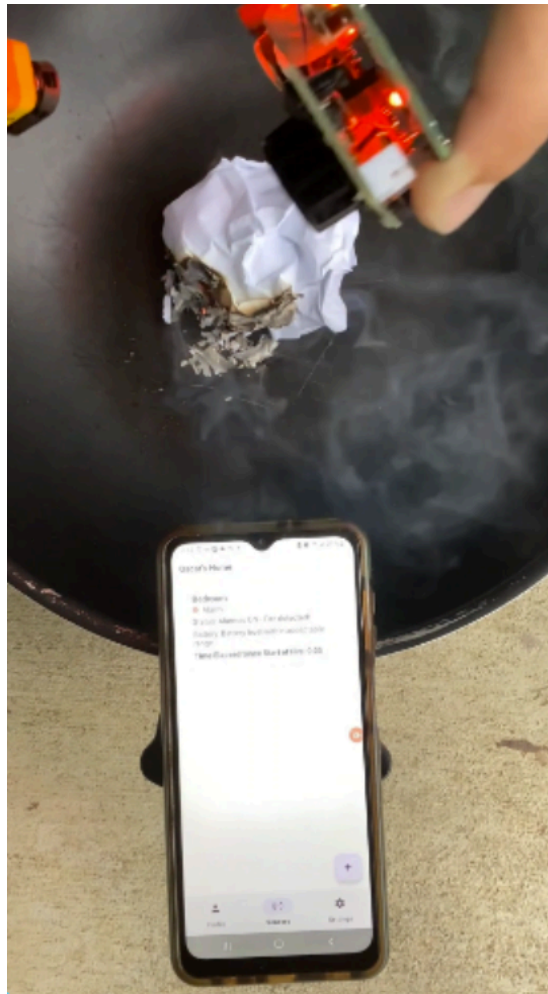


Figure 6: Alarm from smoke threat

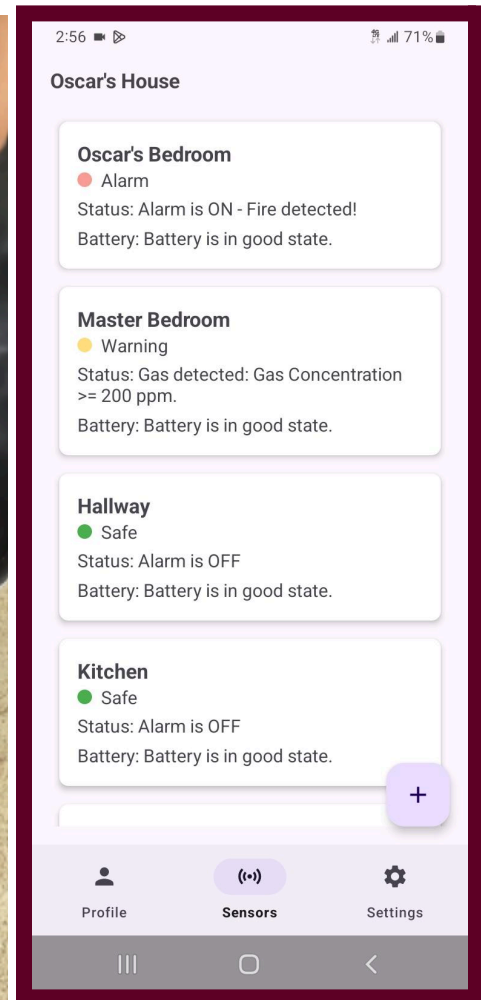


Figure 7: Status per Room

The fire detector was tested in various environments to evaluate its accuracy. As shown in Figure 6, the buzzer consistently sounded when the Pyro Sensor was placed over burning paper. It was also tested against steam from boiling water and a shower, where the alarm did not activate. However, prolonged exposure to cooking smoke triggered the buzzer, indicating the Pyro Sensor should not be placed near cooking areas.

3.5. Power and Wi-Fi Recovery

The Pyro Sensor was tested for backup battery functionality by disconnecting the 120V AC power; the MCU continued scanning peripherals and communicating with Firebase, confirming recovery to the 9V battery. When the router was turned off, the MCU still scanned and triggered

the buzzer locally, though it couldn't send updates to the cloud. A potential solution is using a wired cellular network instead of Wi-Fi.

3.6. Data Flow

The Pyro Sensor mobile application uses real-time listeners connected to the Firebase Realtime Database to monitor sensor status updates. The `addValueEventListener` method continuously listens for changes in the user's sensor data, ensuring that any updates, such as a triggered alarm or status change, are immediately reflected in the application without requiring a manual refresh. This allows for near-instant synchronization between the sensor network and the user's device.

```
// Listen for sensor data changes
database.getReference(path: "users/$uid/sensors")
    .addValueEventListener(object : ValueEventListener {
        override fun onDataChange(snapshot: DataSnapshot) {
            Log.d(TAG, msg: "Sensor data changed. Number of sensors: ${snapshot.children.count()}")

            for (sensorSnapshot in snapshot.children) {
                val sensorId = sensorSnapshot.key ?: continue
                val name = sensorSnapshot.child(path: "Name").child(path: "Sensor").getValue(String::class.java) ?: "Unknown"
                val status = sensorSnapshot.child(path: "Alarm").child(path: "message").getValue(String::class.java) ?: "Unknown"
                val state = sensorSnapshot.child(path: "Alarm").child(path: "alarm_status").getValue(String::class.java) ?: "Safe"

                Log.d(TAG, msg: "Processing sensor: $sensorId")
                Log.d(TAG, msg: "Name: $name, Status: $status, State: $state")
            }
        }
    })
```

Figure 8: Real-time Listeners Code Example

3.7. User Interface Response Time

During testing, the mobile application displayed updated sensor information within approximately < 1 second after a change was detected in the database. Because the app uses real-time listeners, updates such as alarm triggers or changes in sensor state are pushed automatically to the user interface with minimal delay, providing timely awareness of potential fire or smoke events.

3.8. Notification Handling

When an alarm is triggered, the system automatically updates the Firebase database, and a corresponding notification is sent to the user's device through Firebase Cloud Messaging (FCM) and Twilio for SMS alerts to emergency contacts. Notification delivery to the user's phone typically occurs within 1-2 seconds after the sensor triggers, ensuring that users and emergency services are alerted quickly.

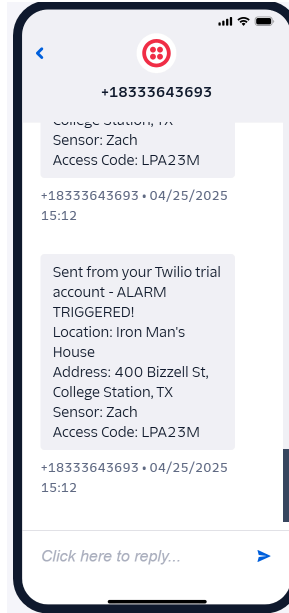


Figure 9: Message Notification Sent to Central Dispatcher

3.9. Data Stored per Sensor

Each sensor monitored through the mobile app stores and displays key information, including the sensor ID, sensor name (user-assigned), alarm message status, and alarm condition ("Safe," "Warning," or "Alarm"). This data structure allows the app to display a clear and concise dashboard of all active sensors and their real-time status across different locations.

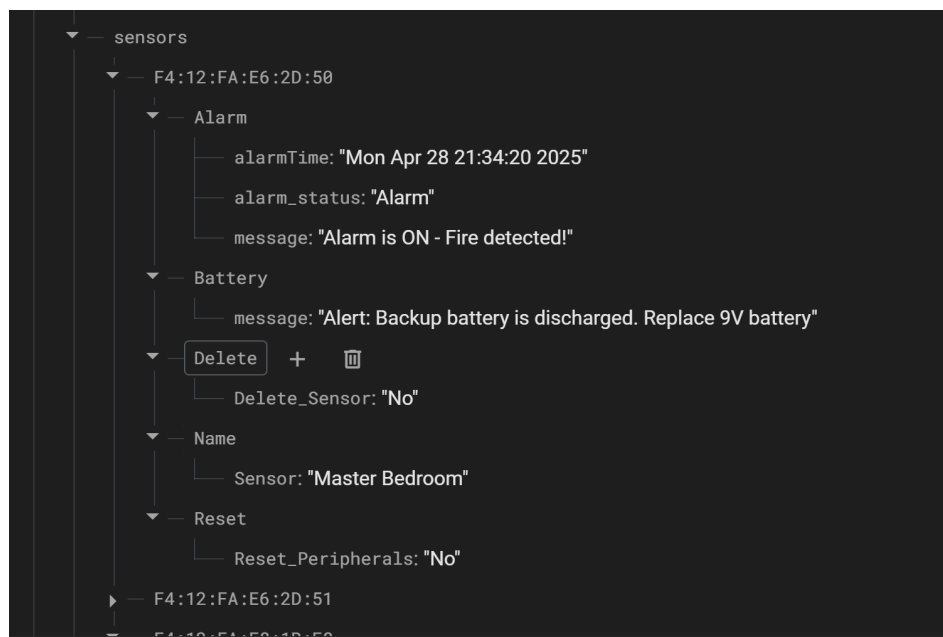


Figure 10: Real-time Database Structure

3.10. Authentication and Security

User access to the mobile application is restricted through Firebase Authentication, requiring users to log in with a registered email and password. Each user's sensor data is isolated within the database under their unique user ID (UID), preventing unauthorized access and ensuring that users only see and manage their own building sensors.

Identifier	Providers	Created ↓	Signed In	User UID
testing@gmail.com		Apr 28, 2025	Apr 28, 2025	QSCsU5z6aeTUPXvsAs1DkRJ...
ironman@gmail.com		Apr 25, 2025	Apr 28, 2025	eAPQ10JnDxZoLnLnMQExaQL...
tonystark1@gmail.com		Apr 25, 2025	Apr 25, 2025	QwJEkemVXFcXWN4GHt7DfO...
testemail77@gmail.com		Apr 24, 2025	Apr 24, 2025	RCPP00w05xRKFCUZbs60xN...
test@gmail.com		Apr 24, 2025	Apr 24, 2025	LqAygkYQ03cNDGtq0PIX23YI...
alfonso.wd17@gmail.c...		Apr 24, 2025	Apr 25, 2025	BD3oNCfnU4X6rmsrs4N59pc...
testemail@gmail.com		Apr 2, 2025	Apr 3, 2025	6nqh1RVRZRZQeliuuaJeDIGQu...
missrev@gmail.com		Mar 3, 2025	Apr 25, 2025	I4liCnEC3sbgS6PJ4NMa7Ae1...

Figure 11: Firebase Authenticated Users

4. Conclusion

The Pyro Sensor System proved to be an effective and reliable solution for improving fire and smoke detection. Through proper research, design, and execution, the project successfully evolved from an initial concept to a fully functional, integrated safety system that enhances emergency response and user accessibility.

4.1. Key Decisions

Early decisions focused on clearly defining the scope and priorities of the Pyro Sensor System. Since the problem statement was given to us — improving fire detection and emergency response — we focused our efforts on designing a system that was affordable, easy to use, and reliable under emergency conditions. Rather than building a traditional alarm system, we decided on a wireless, cloud-connected network of sensors to allow real-time monitoring and notifications.

Choosing Firebase as the cloud platform allowed for fast data updates across devices, and Twilio was selected to automate alert messages directly to dispatchers without relying on user intervention. On the hardware side, we prioritized low-power operation to maximize battery life, and redesigned the sensor board after initial testing revealed issues with power delivery and component integration. Switching from a speaker to a buzzer was another key decision, driven by the ESP32's limited output capabilities, ensuring that alarm sounds could still be triggered reliably without stressing the microcontroller. Each decision was made with the goal of creating a practical, deployable safety system, not just a prototype.

4.2. Lessons Learned

Senior Design provided an invaluable learning experience for our entire team. Taking a complex idea from concept to reality taught us not only technical skills, but also the importance of teamwork, communication, and project management. The final Pyro Sensor System met our original goals, and the process of building it expanded our knowledge across multiple engineering disciplines.

From a technical standpoint, each team member contributed to different aspects of the system and gained important new skills. On the hardware side, we learned how to design, build, and test a custom PCB using Altium, and how to troubleshoot issues during physical integration. We also gained experience programming microcontrollers (ESP32), including setting up reliable Bluetooth Low Energy (BLE) connections, Wi-Fi communication, and processing sensor signals in real time. On the software side, building a full-stack mobile application was a major learning experience. We developed skills in Android development, Firebase real-time database integration, cloud synchronization, and implementing

third-party services like Twilio for automated messaging. The complexity of connecting hardware, cloud services, and a user interface provided real-world experience in full system development.

Another major takeaway was the importance of public speaking and presenting technical material clearly. Regular presentations and feedback sessions helped us become better at delivering concise, professional explanations to a wide range of audiences.

Finally, time management was a critical skill we improved over the course of the project. Coordinating schedules for subsystem integration, testing, and debugging taught us how to plan meetings efficiently and keep focused on the highest priority tasks. The teamwork and organization developed during this project will carry forward into our future professional work.

4.3. Future Work

Several improvements could be made to further develop the Pyro Sensor System into a fully marketable product, including:

- Adding a cellular module to allow communication without relying solely on Wi-Fi
- Implementing user authentication to allow multiple authorized users to monitor and control sensors
- Allowing users to manage multiple buildings or properties under a single account
- Expanding sensor connectivity into a mesh network to improve range and reliability across larger areas
- Improving sensor hardware for faster smoke detection, greater environmental resistance, and longer battery life
- Developing additional hardware fail-safes, such as local backup alarms if cloud communication fails
- Enhancing the mobile app interface to include detailed event history, maintenance reminders, and sensor health monitoring
- Improving system cybersecurity to ensure data privacy and protect against unauthorized access

These improvements would make the Pyro Sensor System more scalable, reliable, and competitive for commercial use.